

Real Analysis Notes

Course Code: MSMAT01DSC04

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1 Basic Topology

1.1 Finite, Countable and Uncountable Sets

Definition 1.1. Let A and B be any two sets. Suppose there is a rule or correspondence that assigns to each element $x \in A$ exactly one element in B , which we denote by $f(x)$. We call this correspondence f a function from A to B , or a mapping of A into B .

In this context, the set A is referred to as the domain of f (meaning f is defined on A), and the individual elements $f(x)$ are the values taken by f . The collection of all such values is called the range of f .

Definition 1.2. Suppose we have a mapping f from a set A into a set B . For any subset $E \subseteq A$, we define $f(E)$ as the set of all elements of the form $f(x)$, where $x \in E$. We refer to $f(E)$ as the image of E under the function f .

Notice that the entire range of f can be written as $f(A)$, and it naturally follows that $f(A) \subseteq B$. In the special case where $f(A) = B$ (meaning every element in the codomain B is mapped to by some element in A), we say that f maps A onto B . Such a function is also commonly called surjective.

Definition 1.3. Let f be a mapping from a set A into a set B . For any subset $E \subseteq B$, we use the notation $f^{-1}(E)$ to denote the set of all elements $x \in A$ which map into E (that is, where $f(x) \in E$). This set $f^{-1}(E)$ is called the inverse image (or preimage) of E under f .

Definition 1.4. For a specific element $y \in B$, $f^{-1}(y)$ represents the set of all $x \in A$ such that $f(x) = y$. If, for every $y \in B$, the set $f^{-1}(y)$ contains at most one element from A , we say that f is a 1-1 (one-to-one) mapping from A into B .

An equivalent and often more practical way to state this is: a function f is 1-1 (or injective) provided that distinct inputs yield distinct outputs. Formally, $f(x_1) \neq f(x_2)$ whenever $x_1 \neq x_2$ for any $x_1, x_2 \in A$.

Definition 1.5. When there exists a 1-1 mapping from a set A onto a set B (a bijection), we say that A and B can be placed in a 1-1 correspondence. We also express this by saying A and B share the same cardinal number, or simply that they are equivalent, denoted by $A \sim B$.

This equivalence behaves as an equivalence relation because it naturally satisfies three key properties:

1. **Reflexivity:** $A \sim A$ for any set A .
2. **Symmetry:** If $A \sim B$, then $B \sim A$.
3. **Transitivity:** If $A \sim B$ and $B \sim C$, then $A \sim C$.

Definition 1.6. Let J denote the set of all positive integers, and for any given positive integer n , let $J_n = \{1, 2, \dots, n\}$ represent the set of the first n positive integers. For an arbitrary set A , we define its size as follows:

- (a) A is finite if there exists some n such that $A \sim J_n$. (By convention, the empty set is also considered finite).
- (b) A is infinite if it is not a finite set.
- (c) A is countable (sometimes referred to as enumerable or denumerable) if $A \sim J$.
- (d) A is uncountable if it is neither finite nor countable.
- (e) A is at most countable if it is either finite or countable.

Remark 1.7. For two finite sets A and B , the equivalence $A \sim B$ clearly holds if and only if A and B contain the exact same number of elements. However, when dealing with infinite sets, simply "counting" elements to establish cardinality does not work, and the intuitive idea of "having the same number of elements" becomes quite vague. This is why the rigorous notion of a 1-1 correspondence is crucial; it allows us to mathematically establish when two infinite sets have the "same size" or "same number of elements" while retaining complete clarity.

Example 1.8. Let A be the set of all integers. We can show that A is countable by setting up a 1-1 correspondence with the set of positive integers J . Consider the following arrangement of the sets A and J :

$$\begin{array}{cccccccc} A: & 0, & 1, & -1, & 2, & -2, & 3, & -3, \dots \\ J: & 1, & 2, & 3, & 4, & 5, & 6, & 7, \dots \end{array}$$

In this case, we can provide an explicit formula for a function f from J to A which establishes this 1-1 correspondence:

$$f(n) = \begin{cases} \frac{n}{2} & \text{if } n \text{ is even,} \\ -\frac{n-1}{2} & \text{if } n \text{ is odd.} \end{cases}$$

Example 1.9. Any two open intervals (a, b) and (c, d) have the same cardinality. We can explicitly construct a bijection $f : (a, b) \rightarrow (c, d)$ using a linear transformation:

$$f(x) = c + \frac{d-c}{b-a}(x-a)$$

This exact same mapping provides a 1-1 correspondence between any two closed intervals $[a, b]$ and $[c, d]$. (Note: For students' understanding, observe that these linear maps are continuous; they simply stretch and shift one interval to perfectly cover the other).

Example 1.10. Any open interval has the same cardinality as the entire set of real numbers $\mathbb{R}$. For instance, consider the open interval $(-\frac{\pi}{2}, \frac{\pi}{2})$ and $\mathbb{R}$. A natural bijection is the tangent function:

$$f(x) = \tan(x) \quad \text{for } x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

(Note: This mapping is also continuous, extending the bounded interval to cover all of $\mathbb{R}$).

Example 1.11. An open interval and a closed interval have the same cardinality, for instance $(0, 1) \sim [0, 1]$. An elegant way to prove this is by using the Schröder-Bernstein Theorem.

Schröder-Bernstein Theorem (without proof): If there exists a 1-1 mapping from a set A into a set B , and a 1-1 mapping from B into A , then there exists a bijection between A and B (i.e., $A \sim B$).

To apply this to $(0, 1)$ and $[0, 1]$, we just need two 1-1 functions:

$$\begin{aligned} f : (0, 1) &\rightarrow [0, 1] \quad \text{defined by } f(x) = x \\ g : [0, 1] &\rightarrow (0, 1) \quad \text{defined by } g(x) = \frac{1}{4} + \frac{1}{2}x \end{aligned}$$

Since both f and g are 1-1, the theorem guarantees $(0, 1) \sim [0, 1]$.

Alternatively, we can construct an explicit bijection. Consider the sequence of points $1/2, 1/3, 1/4, \dots$ inside $(0, 1)$. We can define a function $h : (0, 1) \rightarrow [0, 1]$ that maps the first two terms of this sequence to the endpoints 0 and 1, shifts the rest of the sequence, and leaves everything else unchanged:

$$h(x) = \begin{cases} 0 & \text{if } x = 1/2 \\ 1 & \text{if } x = 1/3 \\ \frac{1}{n-2} & \text{if } x = 1/n \text{ for } n \geq 4 \text{ } (n \in \mathbb{N}) \\ x & \text{if } x \neq 1/n \text{ for any integer } n \geq 2 \end{cases}$$

(Observe that unlike the previous examples, this piecewise explicit bijection is not continuous).

Remark 1.12. In the first two examples, the bijections and their inverses are continuous. You will learn in topology that a continuous bijection with a continuous inverse is called a homeomorphism, which effectively identifies two spaces as having the "same shape".

However, when mapping a closed interval to an open interval (as in our third example), you cannot get a homeomorphism. In fact, no bijection between an open and closed interval can be continuous in both directions. This is due to a topological property you will encounter later: continuous functions map compact sets (like closed, bounded intervals) to compact sets. Since an open interval is not compact, a continuous bijection from a closed interval to an open interval is impossible.

Remark 1.13. Using the sequence-shifting technique or the Schröder-Bernstein theorem from the third example, 1-1 correspondences between open intervals and half-open intervals, or half-open intervals and closed intervals, can be established quite similarly.

The ultimate takeaway from these examples is profound: the cardinality of the real numbers $\mathbb{R}$ is exactly the same as the cardinality of any interval (open, closed, or half-open), provided the interval has a non-zero length.

Remark 1.14. A finite set cannot be equivalent to one of its proper subsets. That this is, however, possible for infinite sets, is demonstrated by our previous example establishing a 1-1 correspondence between the positive integers J and all integers A , even though J is a proper subset of A .

In fact, we could replace our earlier definition of an infinite set with the following statement: A set A is infinite if A is equivalent to one of its proper subsets.

Definition 1.15. By a sequence, we mean a function f defined on the set J of all positive integers. If we evaluate this function at an integer $n \in J$ and let $f(n) = x_n$, it is customary to denote the sequence by the symbol $\{x_n\}$, or simply by listing its outputs: $x_1, x_2, x_3, \dots$.

The values of f , that is, the elements x_n , are called the terms of the sequence. If A is a set and if $x_n \in A$ for all $n \in J$, then $\{x_n\}$ is said to be a sequence in A , or a sequence of elements of A . Note that unlike sets, the terms $x_1, x_2, x_3, \dots$ of a sequence need not be distinct.

Remark 1.16. The concept of a sequence gives us a highly intuitive way to understand countable sets. Since a countable set has a 1-1 correspondence with the positive integers J , every countable set can be viewed as the range of a sequence where all terms are distinct.

In simple words, if a set is countable, its elements can be enumerated. This means they can be "arranged in a sequence." You can literally line the elements up and label them: "Here is the first element (x_1), here is the second element (x_2), here is the third (x_3)," and so on. You are guaranteed that every single element in the set will eventually appear on your list exactly once.

(As a minor note, sometimes it is more convenient to replace J in our definition with the set of all non-negative integers, meaning we simply start our sequence index with 0 rather than 1).

Theorem 1.17. Every infinite subset of a countable set A is countable.

Proof. Suppose $E \subset A$, and E is an infinite set. Our goal is to show that we can arrange the elements of E into a sequence, proving it is countable.

Step 1: List the elements of the parent set. Since A is countable, we know we can arrange all of its elements into a single sequence of distinct terms: $x_1, x_2, x_3, \dots$.

Step 2: Pick out the elements of E in order. We will build a sequence just for E by scanning through our list of A and picking out the elements that belong to E . We do this by keeping track of their index positions:

- Let n_1 be the smallest positive integer such that $x_{n_1} \in E$. This is the first element of E we encounter.
- Having found the first element, let n_2 be the next smallest integer (greater than n_1) such that $x_{n_2} \in E$.
- In general, once we have found $k-1$ elements (at positions $n_1, \dots, n_{k-1}$), we find the next element by letting n_k be the smallest integer greater than n_{k-1} such that $x_{n_k} \in E$.

Step 3: Establish the 1-1 correspondence. Because E is an infinite set, this scanning process will never stop; we will never run out of elements. We can define a function f from the set of positive integers J to E by $f(k) = x_{n_k}$ for $k = 1, 2, 3, \dots$.

This function matches every positive integer k to exactly one distinct element of E , exhausting all elements of E in the process. This establishes a perfect 1-1 correspondence between J and E , meaning E is countable. $\square$

Remark 1.18. This theorem reveals a fundamental property of infinite sets: roughly speaking, countable sets represent the "smallest" type of infinity. Because any infinite subset of a countable set simply drops down to being countable again, it is impossible for an uncountable set to exist as a subset of a countable set.

Definition 1.19. Let A and Ω be sets, and suppose that with each element α of A there is associated a subset of Ω which we denote by E_α . The set whose elements are these subsets E_α is denoted by $\{E_\alpha\}$. Instead of confusingly saying "a set of sets," we generally refer to this as a collection of sets or a family of sets.

For any such family of sets, we can generalize the operations of union and intersection:

- **Arbitrary Union:** The union of the sets E_α is defined to be the set S such that $x \in S$ if and only if $x \in E_\alpha$ for at least one $\alpha \in A$. We use the notation:

$$S = \bigcup_{\alpha \in A} E_\alpha$$

- **Arbitrary Intersection:** The intersection of the sets E_α is defined to be the set P such that $x \in P$ if and only if $x \in E_\alpha$ for every $\alpha \in A$. We use the notation:

$$P = \bigcap_{\alpha \in A} E_\alpha$$

Finally, if $A \cap B$ is not empty, we say that A and B intersect; if it is empty, they are disjoint.

Remark 1.20. To make sense of this notation, it helps to think of the set A as an "index set" (like a collection of name tags). The way we write unions and intersections changes depending on how large this index set is:

- **Finite Collections:** If A just consists of the integers $1, 2, \dots, n$, we are dealing with a finite number of sets. We usually write the union explicitly as $S = E_1 \cup E_2 \cup \dots \cup E_n$, or compactly using a limit: $S = \bigcup_{m=1}^n E_m$. The exact same logic and notation apply to finite intersections, replacing the $\cup$ with $\cap$.
- **Countable Infinite Collections:** If A is the set of all positive integers, we have an infinite sequence of sets. The usual notation for their union becomes $S = \bigcup_{m=1}^{\infty} E_m$. It is important to note that the ∞ symbol here simply indicates that we are taking the union of a countable collection of sets; it should not be confused with the literal numbers $+\infty$ or $-\infty$.
- **Uncountable Arbitrary Collections:** The generalized notation $\bigcup_{\alpha \in A} E_\alpha$ is incredibly powerful because it works even if A is an uncountable set (like the real numbers). It guarantees that no matter how unimaginably large the family of sets is, we can rigorously define what it means to combine them all together or find what they all share.

Example 1.21. Let A be the set of real numbers x such that $0 < x \leq 1$. We can write this index set as $A = (0, 1]$.

For every $x \in A$, let E_x be the set of real numbers y such that $0 < y < x$. In other words, each set E_x is the open interval $(0, x)$.

Then this family of sets has the following properties:

(i) $E_x \subset E_z$ if and only if $0 < x \leq z \leq 1$;

(ii) $\bigcup_{x \in A} E_x = E_1$;

(iii) $\bigcap_{x \in A} E_x$ is empty.

Explanation of Properties:

- **Property (i)** is visually clear. The open interval $(0, x)$ is completely contained inside the interval $(0, z)$ if and only if the upper bound x is less than or equal to z . This shows our sets are "nested" inside one another.

- **Property (ii)** is also straightforward. Since the sets are nested, as x gets larger, the interval E_x gets larger. The union of all these intervals is simply the largest interval in the collection. Because x can take a maximum value of 1 (since $x \in A$), the largest set is $E_1 = (0, 1)$. Every element in the union will fall into this maximal set.
- **To prove Property (iii)**, we must show that no single real number belongs to every set E_x . Suppose, for the sake of contradiction, that there is some real number y in the intersection. For y to be in any E_x , it must be strictly positive ($y > 0$).

However, no matter how tiny this positive number y is, we can always find a real number $x \in A$ that is strictly smaller than y (for instance, let $x = y/2$). For this specific choice of x , y is not in E_x because y is not less than x . Since $y \notin E_x$ for this particular x , it cannot be in the intersection of all E_x . Thus, the intersection must be empty.

Theorem 1.22. Let $\{E_n\}$, $n = 1, 2, 3, \dots$, be a sequence of countable sets, and put

$$S = \bigcup_{n=1}^{\infty} E_n.$$

Then S is countable.

Before diving into the formal proof, it is helpful to state what this theorem means in simple terms: **a countable union of countable sets is countable.**

Proof. Since each individual set E_n is countable, its elements can be arranged in a sequence. Let us denote the elements of the first set E_1 as $x_{11}, x_{12}, x_{13}, \dots$, the elements of the second set E_2 as $x_{21}, x_{22}, x_{23}, \dots$, and in general, the elements of the n -th set E_n as $\{x_{nk}\}$ for $k = 1, 2, 3, \dots$.

Step 1: Construct an infinite grid.

We can arrange all the elements of all the sets into an infinite two-dimensional array, where the n -th row contains all the elements of the set E_n :

$$\begin{array}{ccccccc} x_{11} & x_{12} & x_{13} & x_{14} & \dots & & \\ x_{21} & x_{22} & x_{23} & x_{24} & \dots & & \\ x_{31} & x_{32} & x_{33} & x_{34} & \dots & & \\ x_{41} & x_{42} & x_{43} & x_{44} & \dots & & \\ \vdots & \vdots & \vdots & \vdots & \ddots & & \end{array}$$

This grid contains every single element of the union set S .

Step 2: Traverse the grid to form a single sequence.

To show that S is countable, we must show that all elements in this grid can be listed in a single, one-dimensional sequence without missing any. We cannot just read across the first row, because it is infinitely long and we would never reach the second row.

Instead, we read the grid by tracing finite diagonals (moving from bottom-left to top-right along diagonals). This gives us the sequence:

$$x_{11}; \quad x_{21}, x_{12}; \quad x_{31}, x_{22}, x_{13}; \quad x_{41}, x_{32}, x_{23}, x_{14}; \quad \dots$$

Step 3: Address duplicate elements.

The sets E_n might not be mutually disjoint; they might share elements. If they do, some elements will appear more than once in our new diagonal sequence.

If we move through our sequence and simply skip any element we have already listed, we create a 1-1 correspondence between the unique elements of S and a subset (let's call it T) of the positive integers. This establishes that S is *at most countable*.

Step 4: Conclude using previous results.

We know that the first set E_1 is completely contained within S (that is, $E_1 \subset S$). Since E_1 is an infinite set, S must also be an infinite set.

From Step 3, S is equivalent to T , which is a subset of the positive integers. Since S is infinite, T must be an *infinite* subset of the positive integers. By our previous theorem, **every infinite subset of a countable set is countable**. Because T is an infinite subset of the countable set of positive integers, T is countable, and therefore S is countable. $\square$

Corollary 1.22.1. *Suppose A is at most countable, and, for every $\alpha \in A$, B_α is at most countable. Put*

$$T = \bigcup_{\alpha \in A} B_\alpha.$$

Then T is at most countable.

Proof. The text provides a very compact proof for this, stating that T is equivalent to a subset of the previously defined set S from equation 15. Here is how that deduction works.

The previous theorem proved that a strictly countable union of strictly countable sets is countable. This corollary relaxes those conditions to "at most countable," meaning the index set A might be finite, and the individual sets B_α might also be finite or even empty.

Because A is at most countable, we can arrange its elements into a sequence $\alpha_1, \alpha_2, \dots$. This allows us to list our sets as $B_{\alpha_1}, B_{\alpha_2}, \dots$. If A is finite, this list eventually stops. We can easily turn it into an infinite sequence by simply defining all the remaining sets in the sequence as empty sets. Adding empty sets to a union does not change the final set T .

Similarly, because each B_α is at most countable, it is either equivalent to a fully countable set or it is a subset of one. Therefore, when we take the union of all these B_α sets to form T , we are essentially building a set that is contained entirely within the union S from equation 15.

Because the previous theorem definitively proved that the larger set S is countable, and T is merely a subset of S , it logically follows that T must be at most countable. $\square$

Theorem 1.23. *Let A be a countable set, and let B_n be the set of all n -tuples $(a_1, \dots, a_n)$, where $a_k \in A$ ($k = 1, \dots, n$), and the elements $a_1, \dots, a_n$ need not be distinct. Then B_n is countable.*

Proof. We can prove this theorem using the principle of mathematical induction.

First, consider the base case where $n = 1$. The set B_1 consists of all 1-tuples, which are simply the individual elements of A . Since B_1 is exactly the same set as A , and we are given that A is countable, it is evident that B_1 must also be countable.

Next, we move to the inductive step. We assume that the theorem holds true for some integer $n - 1$. This means we assume the set B_{n-1} , which contains all $(n - 1)$ -tuples, is a countable set. We must show that this assumption implies the set B_n is also countable.

Any n -tuple in B_n can be constructed by taking an $(n - 1)$ -tuple and appending one more element to it. Therefore, an element in B_n can be written in the form (b, a) , where b represents an entire $(n - 1)$ -tuple from the set B_{n-1} , and a is a single element from our original set A .

Let us fix one specific $(n - 1)$ -tuple, which we will call b . We can form a collection of pairs by keeping this b constant and pairing it with every possible element a from the set A . Because the set A is countable, this specific collection of pairs (b, a) is equivalent in size to A , making it a countable set itself.

To construct the entire set B_n , we must perform this pairing process for every single $(n - 1)$ -tuple b that exists in B_{n-1} . We are essentially taking a union of all these individual, countable collections of pairs. The number of these collections is determined by the number of elements in B_{n-1} . Since our induction hypothesis established that B_{n-1} is a countable set, we are combining a countable number of these collections.

Consequently, B_n is formed by taking a countable union of countable sets. As established by our previous theorem, a countable union of countable sets is always countable. Thus, B_n is countable, and the proof follows by induction. $\square$

Corollary 1.23.1. *The set of all rational numbers is countable.*

Proof. The text provides a concise proof by referencing our previous theorem about n -tuples. Here is how that connection works in detail.

Recall that a rational number is defined as any number that can be expressed as a fraction b/a , where both b and a are integers (with a not equal to zero). This means every single rational number can be represented by a pair of integers (b, a) .

Our previous theorem established that if a base set is countable, then the set of all n -tuples formed from that base set is also countable. If we let our base set be the set of all integers, which we already proved is a countable set, and we set $n = 2$, the theorem guarantees that the set of all possible integer pairs (b, a) is countable.

Because every rational number is formed by one of these integer pairs, the set of all fractions b/a is essentially a subset of this larger set of pairs. We simply discard the pairs where the denominator

a is zero, and we recognize that some pairs represent the same fraction (like $1/2$ and $2/4$). Since the entire collection of integer pairs is countable, this sub-collection of valid fractions must also be countable. Therefore, the set of all rational numbers is countable. $\square$

Exercise 1. *Prove that the set of all algebraic numbers is countable.*

Proof. Before proving the statement, we must first define what an algebraic number is. An algebraic number is any complex number x that is a root of a non-zero polynomial equation with integer coefficients. In other words, x is a solution to an equation of the form

$$a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0 = 0$$

where the degree $n \geq 1$, all the coefficients $a_0, a_1, \dots, a_n$ are integers, and the leading coefficient a_n is not zero.

To prove that the set of all such numbers is countable, we can organize them by the degree n of their corresponding polynomials. Let us fix a specific degree n . A polynomial of this degree is completely determined by its coefficients, which we can list as an $(n+1)$ -tuple of integers: $(a_0, a_1, \dots, a_n)$.

Because the set of all integers is countable, our earlier theorem regarding tuples guarantees that the set of all possible $(n+1)$ -tuples of integers is also countable. Therefore, there are only countably many polynomials of degree n with integer coefficients.

Furthermore, a fundamental theorem of algebra states that any polynomial of degree n has at most n distinct roots. This means each polynomial only yields a finite set of algebraic numbers. To gather all algebraic numbers associated with degree n , we take the union of the roots from each of these polynomials. Since we have a countable number of polynomials, and each has a finite number of roots, we are taking a countable union of finite sets. This guarantees that the set of all algebraic numbers of a fixed degree n is countable.

Finally, to construct the entire set of all algebraic numbers, we must combine the sets of roots for every possible degree $n = 1, 2, 3, \dots$. Since the set of algebraic numbers for each individual degree n is countable, combining all of them together means we are taking a countable union of countable sets. As established by our previous theorem, a countable union of countable sets is always countable. Thus, the set of all algebraic numbers is countable. $\square$

Theorem 1.24. *Let A be the set of all sequences whose elements are the digits 0 and 1. This set A is uncountable. (Note: The elements of A are sequences like $1, 0, 0, 1, 0, 1, 1, 1, \dots$).*

Proof. We will prove this by showing that no countable list can possibly contain every sequence in A .

Step 1: Assume a countable subset.

Let E be any countable subset of A . Because E is countable, we can enumerate its elements in a sequence: $s_1, s_2, s_3, \dots$. Keep in mind that each element s_n in this list is itself an infinite sequence of 0s and 1s.

Step 2: Construct a new sequence (Cantor's Diagonal Process).

We will now construct a brand new sequence, which we will call s . We determine the digits of s one by one using a famous technique known as **Cantor's diagonal process**:

- Look at the 1st digit of the first sequence s_1 . If it is a 1, make the 1st digit of s a 0. If it is a 0, make it a 1.
- Look at the 2nd digit of the second sequence s_2 . If it is a 1, make the 2nd digit of s a 0, and vice versa.
- In general, for any integer n , look at the n -th digit of the sequence s_n . If that digit is a 1, we define the n -th digit of our new sequence s to be 0, and vice versa.

Step 3: Show the new sequence is missing from the list.

Because our new sequence s consists entirely of 0s and 1s, it is clearly a valid element of the overall set A . But is it in our countable subset E ? By our construction, s differs from the first sequence s_1 in the first digit. It differs from the second sequence s_2 in the second digit. In general, it differs from every sequence s_n in our list at the n -th digit. Therefore, s is not anywhere in the set E .

Step 4: Conclusion.

Since $s \in A$ but $s \notin E$, it means E is missing at least one element of A . Thus, E is a *proper subset* of A .

We have just shown that *every* countable subset of A is merely a proper subset of A . It logically follows that A itself must be uncountable. (If A were countable, we could let our list E be the entirety of A . But that would mean A is a proper subset of itself, which is absurd). $\square$

Exercise 2. Prove that the set of all real numbers $\mathbb{R}$ is uncountable.

Proof. We will prove this by contradiction, relying on a principle established earlier: every infinite subset of a countable set must itself be countable.

Assume, for the sake of contradiction, that the entire set of real numbers $\mathbb{R}$ is countable.

Let us look at a very specific subset of the real numbers. Let A be the set of all real numbers between 0 and 1 whose decimal representations consist exclusively of the digits 0 and 1 (for example, numbers like 0.1011001... or 0.0101010...). Clearly, A is a subset of $\mathbb{R}$.

Notice that there is a perfect 1-1 correspondence between the numbers in set A and the set of all infinite sequences made up of the digits 0 and 1. We proved in our previous theorem (using Cantor's diagonal process) that the set of all such 0 and 1 sequences is uncountable. Because they map perfectly to one another, our subset A must also be an uncountable set.

This creates a fatal logical flaw. Under our assumption that $\mathbb{R}$ is countable, every infinite subset of $\mathbb{R}$ must also be countable. However, we just constructed a subset A that is strictly uncountable. This contradiction proves that our initial assumption was false.

Therefore, the set of all real numbers $\mathbb{R}$ is uncountable. $\square$

Remark 1.25. This profound result leads to several important consequences regarding the "size" of different collections of numbers on the real line.

Every interval is uncountable: It is a known mathematical fact that any continuous interval of real numbers (such as the open interval $(0, 1)$ or any interval (a, b)) has the exact same cardinality as the entire set of real numbers $\mathbb{R}$. Since we just proved that $\mathbb{R}$ is uncountable, it immediately shows that every interval on the number line is also uncountable.

The irrational numbers are uncountable: The set of real numbers $\mathbb{R}$ is simply the union of the rational numbers (which we denote as $\mathbb{Q}$) and the irrational numbers. We have already proven that the rational numbers $\mathbb{Q}$ are countable.

If the set of irrational numbers were also countable, then $\mathbb{R}$ would be formed by taking the union of two countable sets. We know that a union of countable sets is always countable, which would mean $\mathbb{R}$ is countable. But $\mathbb{R}$ is uncountable! The only way to avoid this contradiction is if the set of irrational numbers is uncountable.

Intervals are dominated by irrationals: We can apply this exact same logic to any specific segment of the number line. Take any interval I . The rational numbers that happen to fall inside this interval, denoted as $I \cap \mathbb{Q}$, simply form a subset of all rational numbers, meaning they are countable.

The rest of the numbers in the interval are irrational. If the irrationals in the interval were countable, the entire interval I would be the union of two countable sets, making the interval itself countable. Since we know every interval is uncountable, its irrational part must be uncountable. In simple terms, within any interval, the rational numbers are merely a countable dusting, while the irrational numbers make up the uncountable bulk.

1.2 Metric Spaces

Definition 1.26. A set X , whose elements we shall call points, is said to be a metric space if with any two points p and q of X there is associated a real number $d(p, q)$, called the distance from p to q , such that

- (a) $d(p, q) > 0$ if $p \neq q$; $d(p, p) = 0$;
- (b) $d(p, q) = d(q, p)$;
- (c) $d(p, q) \leq d(p, r) + d(r, q)$, for any $r \in X$.

Any function with these three properties is called a distance function, or a metric.

Example 1.27. The most important examples of metric spaces, from our standpoint, are the euclidean spaces $\mathbb{R}^k$, especially $\mathbb{R}^1$ (the real line) and $\mathbb{R}^2$ (the complex plane); the distance in $\mathbb{R}^k$ is defined by

$$d(\mathbf{x}, \mathbf{y}) = |\mathbf{x} - \mathbf{y}| \quad (\mathbf{x}, \mathbf{y} \in \mathbb{R}^k).$$

By Theorem 1.37, the conditions of Definition 2.15 are satisfied by this formula.

Remark 1.28. In Rudin, an open interval (a, b) in $\mathbb{R}^1$ is called a segment, while the term interval is reserved for a closed interval $[a, b]$. Throughout these notes, we explicitly refer to them as open intervals, closed intervals, and half-open intervals.

Definition 1.29. If $a_i < b_i$ for $i = 1, \dots, k$, the set of all points $\mathbf{x} = (x_1, \dots, x_k) \in \mathbb{R}^k$ whose coordinates satisfy the inequalities $a_i \leq x_i \leq b_i$ ($1 \leq i \leq k$) is called a k -cell. Thus, a 1-cell is a closed interval, a 2-cell is a rectangle, etc.

Definition 1.30. If $\mathbf{x} \in \mathbb{R}^k$ and $r > 0$, the open ball (or simply ball) with center at $\mathbf{x}$ and radius r is defined to be the set of all $\mathbf{y} \in \mathbb{R}^k$ such that $|\mathbf{y} - \mathbf{x}| < r$.

Similarly, the closed ball with center at $\mathbf{x}$ and radius r is defined as the set of all $\mathbf{y} \in \mathbb{R}^k$ such that $|\mathbf{y} - \mathbf{x}| \leq r$.

Definition 1.31. A set $E \subset \mathbb{R}^k$ is said to be convex if

$$\lambda \mathbf{x} + (1 - \lambda) \mathbf{y} \in E$$

whenever $\mathbf{x} \in E$, $\mathbf{y} \in E$, and $0 < \lambda < 1$.

Example 1.32. Open and closed balls in $\mathbb{R}^k$ are convex.

To prove this for an open ball B centered at $\mathbf{x}$ with radius r , let $\mathbf{y}, \mathbf{z} \in B$ and choose $0 < \lambda < 1$. By definition of the open ball, we have $|\mathbf{y} - \mathbf{x}| < r$ and $|\mathbf{z} - \mathbf{x}| < r$. Applying the triangle inequality yields:

$$\begin{aligned} |\lambda \mathbf{y} + (1 - \lambda) \mathbf{z} - \mathbf{x}| &= |\lambda(\mathbf{y} - \mathbf{x}) + (1 - \lambda)(\mathbf{z} - \mathbf{x})| \\ &\leq \lambda |\mathbf{y} - \mathbf{x}| + (1 - \lambda) |\mathbf{z} - \mathbf{x}| \\ &< \lambda r + (1 - \lambda) r \\ &= r. \end{aligned}$$

Since $|\lambda \mathbf{y} + (1 - \lambda) \mathbf{z} - \mathbf{x}| < r$, the point $\lambda \mathbf{y} + (1 - \lambda) \mathbf{z}$ lies inside B , proving B is convex.

The exact same proof applies to closed balls (replacing strict inequalities with $\leq$). Additionally, k -cells are also convex.

Definition 1.33. Let X be a metric space. All points and sets mentioned below are understood to be elements and subsets of X .

- (a) A neighborhood of p is a set $N_r(p)$ consisting of all q such that $d(p, q) < r$, for some $r > 0$. The number r is called the radius of $N_r(p)$.
- (b) A point p is a limit point of the set E if every neighborhood of p contains a point $q \neq p$ such that $q \in E$.
- (c) If $p \in E$ and p is not a limit point of E , then p is called an isolated point of E .
- (d) E is closed if every limit point of E is a point of E .
- (e) A point p is an interior point of E if there is a neighborhood N of p such that $N \subset E$.
- (f) E is open if every point of E is an interior point of E .
- (g) The complement of E (denoted by E^c) is the set of all points $p \in X$ such that $p \notin E$.
- (h) E is perfect if E is closed and if every point of E is a limit point of E .
- (i) E is bounded if there is a real number M and a point $q \in X$ such that $d(p, q) < M$ for all $p \in E$.
- (j) E is dense in X if every point of X is a limit point of E , or a point of E (or both).

Example 1.34 (Neighborhoods). Consider the set of real numbers $\mathbb{R}$ with the standard metric $d(x, y) = |x - y|$. An ϵ -neighborhood of a point $x \in \mathbb{R}$ is exactly the open interval $(x - \epsilon, x + \epsilon)$.

Similarly, in $\mathbb{R}^k$ with the standard Euclidean metric, neighborhoods of a point $\mathbf{x}$ are open balls centered at $\mathbf{x}$. However, the "shape" of a neighborhood depends on the chosen metric. For instance, consider $\mathbb{R}^2$ equipped with the maximum metric $d(\mathbf{x}, \mathbf{y}) = \max(|x_1 - y_1|, |x_2 - y_2|)$. In this metric space, a neighborhood (an open ball) takes the geometric shape of an open square with sides parallel to the coordinate axes.

The space itself also restricts neighborhoods. Let the metric space be $X = (0, 2]$ with the standard Euclidean distance. The r -neighborhood of the point 2 is defined relative to X , meaning $N_r(2) = (2 - r, 2 + r) \cap X$. For example, if $r = 1$, the neighborhood is $N_1(2) = (1, 3) \cap (0, 2] = (1, 2]$.

Example 1.35 (Limit Points). *A limit point of a set E does not actually need to belong to the set E . Consider $E = (0, 2]$ as a subset of the metric space $\mathbb{R}$. Every element of E is a limit point of E . Furthermore, 0 is also a limit point of E because every open interval around 0 will intersect $(0, 2]$, even though $0 \notin E$.*

Now, change the context: suppose we consider $X = (0, 2]$ itself as our entire metric space. The candidates for limit points of any subset must come from the metric space X under consideration. In this scenario, 0 cannot be a limit point of any subset of X simply because 0 does not exist in the space X .

Example 1.36 (Isolated Points). *Consider the set $E = (0, 2] \cup \{3\}$ as a subset of $\mathbb{R}$ with the standard metric. The point 3 is an isolated point of E because we can find a neighborhood around 3 (e.g., radius $r = 1/2$) that contains no other points of E .*

If we take the natural numbers $\mathbb{N}$ as a subset of $\mathbb{R}$, every single point in $\mathbb{N}$ is an isolated point, and $\mathbb{N}$ has absolutely no limit points in $\mathbb{R}$.

Consider the set $S = \mathbb{Q} \times (\mathbb{R} \setminus \mathbb{Q})$ (rational x -coordinates and irrational y -coordinates) as a subset of $\mathbb{R}^2$ with the standard metric. Does S have any isolated points? No. Every neighborhood of any point in S contains infinitely many other points of S . Its set of limit points is the entirety of $\mathbb{R}^2$.

Example 1.37 (Closed Sets). *The set of natural numbers $\mathbb{N}$ is closed in $\mathbb{R}$. By definition, a set is closed if it contains all of its limit points. Since $\mathbb{N}$ has no limit points, the condition is vacuously satisfied.*

The entire space $\mathbb{R}$ is closed in $\mathbb{R}$.

The set of rational numbers $\mathbb{Q}$ is not closed in $\mathbb{R}$. Between any two real numbers there exists a rational number, meaning we can construct a sequence of rationals converging to any irrational number. Thus, every irrational number is a limit point of $\mathbb{Q}$, but these limit points are not contained in $\mathbb{Q}$.

Example 1.38 (Interior Points and Open Sets). *Let the metric space be $\mathbb{R}$ and consider the subset $E = (0, 2]$. Every point $x \in (0, 2)$ is an interior point. To rigorously see this, for any $x \in (0, 2)$, choose a radius $r = \min(x, 2 - x)$. The neighborhood $N_r(x)$ will lie entirely inside $(0, 2]$. However, the point 2 is not an interior point because any neighborhood $(2 - r, 2 + r)$ in $\mathbb{R}$ will contain points strictly greater than 2, which fall outside E .*

Now, consider the metric space $X = (0, 2]$ with the standard metric, and let $E = (1, 2]$ be a subset of X . In this relative context, 2 is an interior point of E . A neighborhood of 2 in X with radius 1 is $(1, 2]$, which is entirely contained in E . Consequently, while $(1, 2]$ is not an open set in $\mathbb{R}$, it is an open set in the metric space $X = (0, 2]$.

In general, within $\mathbb{R}$, open intervals are the most fundamental examples of open sets. Furthermore, any union of open intervals is again an open set.

Example 1.39 (Perfect Sets). *Perfect sets are closed sets that have no isolated points (i.e., every point in the set is a limit point of the set). Standard examples in $\mathbb{R}^n$ include closed intervals (or cells) and finite unions of non-degenerate closed intervals. We will explore more intricate examples of perfect sets (such as the Cantor set) later in the course.*

Remark 1.40 (Bounded Sets). *By definition, a set E in a metric space X is bounded if we can find a neighborhood of some point $q \in X$ with a finite radius M that completely contains E . If we are working in $\mathbb{R}$ and we conventionally choose the center point $q = 0$, this translates to $E \subset (-M, M)$, meaning $|x| < M$ for all $x \in E$. This perfectly recovers the standard definition of bounded subsets of real numbers you studied in your previous undergraduate courses.*

Example 1.41 (Dense Sets). *A set E is dense in X if every point of X is either a point of E or a limit point of E (or both). Equivalently, every non-empty open set in X intersects E .*

Consider the metric space $X = (0, 2] \cup \{3\}$ with the standard metric. Let E be the union of the irrationals in $(0, 2]$ and the singleton set $\{3\}$. E is dense in X . Notice that $3 \in E$, but 3 is not a limit point of E (it is an isolated point). This perfectly illustrates why the definition states "is a limit point or a point of E ."

As another example, the set $\mathbb{Q} \times (\mathbb{R} \setminus \mathbb{Q})$ is dense in $\mathbb{R}^2$ with the standard metric, because every open ball in $\mathbb{R}^2$, no matter how small, will contain points with rational x -coordinates and irrational y -coordinates.

Theorem 1.42. *Every neighborhood is an open set.*

Proof. Let $E = N_r(p)$ be a neighborhood of a point p with radius $r > 0$. To prove that E is an open set, we must show that every point inside E is an interior point.

Step 1: Pick an arbitrary point.

Let q be any point in E . By the definition of a neighborhood, the distance from p to q is strictly less than r , meaning $d(p, q) < r$.

Step 2: Find a suitable radius around q .

Because $d(p, q) < r$, there is some "leftover" distance between q and the "boundary" of our neighborhood E . Let us define this positive leftover distance as a real number h :

$$h = r - d(p, q)$$

Notice that $h > 0$ (since $d(p, q) = r - h$). We claim that the neighborhood of q with radius h , denoted $N_h(q)$, is completely contained inside our original set E .

Step 3: Verify using the triangle inequality.

Take any point $s \in N_h(q)$. By definition, this means $d(q, s) < h$. We want to determine if s is also inside E , which requires checking its distance from the original center p .

Using the triangle inequality, we evaluate the distance between p and s :

$$d(p, s) \leq d(p, q) + d(q, s)$$

Since we know $d(q, s) < h$, we can strictly bound this sum:

$$d(p, s) < d(p, q) + h$$

Now, substitute our earlier definition of h into the inequality:

$$d(p, s) < d(p, q) + (r - d(p, q)) = r$$

Step 4: Conclusion.

Since $d(p, s) < r$, the point s must belong to E . Because this is true for every point $s \in N_h(q)$, we have successfully shown that an entire neighborhood around q is contained in E .

Thus, q is an interior point of E . Since q was chosen arbitrarily, every point of E is an interior point, proving that the neighborhood E is an open set. $\square$

Theorem 1.43. *If p is a limit point of a set E , then every neighborhood of p contains infinitely many points of E .*

Proof. We will prove this theorem by contradiction, breaking it down logically.

Step 1: Assume the contrary.

Suppose p is a limit point of E , but there exists some neighborhood N of p which contains only a finite number of points of E .

Step 2: Isolate the finite points.

Let $q_1, \dots, q_n$ be those specific points in the intersection $N \cap E$ which are distinct from p .

Step 3: Construct a smaller neighborhood.

We evaluate the distance from our point p to each of these finitely many points q_m . Let r be the absolute smallest of these distances:

$$r = \min_{1 \leq m \leq n} d(p, q_m)$$

Because we are taking the minimum of a finite set of strictly positive numbers (since $q_m \neq p$, the distance to p is greater than zero), the minimum itself must be strictly positive. Thus, $r > 0$.

Step 4: Reach a contradiction.

Now consider the new, smaller neighborhood $N_r(p)$ centered at p with this radius r . Because r is less than or equal to the distance to any of the points $q_1, \dots, q_n$, this new neighborhood $N_r(p)$ contains absolutely no point q of E such that $q \neq p$.

However, the fundamental definition of a limit point demands that *every* neighborhood of p must contain at least one point of E distinct from p . We have just explicitly constructed a neighborhood $N_r(p)$ that fails this requirement. This contradiction establishes the theorem. $\square$

Corollary 1.43.1. *A finite set has no limit points.*

Proof. By the preceding theorem, if a point p were a limit point of a set E , every neighborhood of p would be required to contain infinitely many distinct points of E . However, if E is finite, no neighborhood can possibly contain infinitely many points of E . Therefore, a finite set cannot have any limit points. $\square$

Theorem 1.44. Let $\{E_\alpha\}$ be a (finite or infinite) collection of sets E_α . Then

$$\left(\bigcup_{\alpha} E_{\alpha}\right)^c = \bigcap_{\alpha} (E_{\alpha}^c).$$

Proof. To prove that these two sets are equal, we must show mutual inclusion: the set on the left is a subset of the set on the right, and conversely, the set on the right is a subset of the set on the left.

Let $A = (\bigcup_{\alpha} E_{\alpha})^c$ and let $B = \bigcap_{\alpha} (E_{\alpha}^c)$.

Step 1: Show that $A \subset B$.

Suppose x is an arbitrary element in A . By the definition of the complement, this means x does not belong to the union $\bigcup_{\alpha} E_{\alpha}$. For an element to be excluded from the union of a collection of sets, it must be absent from every single set in that collection. Therefore, $x \notin E_{\alpha}$ for any α . This implies that x belongs to the complement of each individual set, meaning $x \in E_{\alpha}^c$ for every α . Because x is in every such complement, it must belong to their intersection. Thus, $x \in \bigcap_{\alpha} (E_{\alpha}^c)$, which means $x \in B$. This establishes that $A \subset B$.

Step 2: Show that $B \subset A$.

Conversely, suppose x is an arbitrary element in B . This means $x \in \bigcap_{\alpha} (E_{\alpha}^c)$. By the definition of an intersection, x must belong to E_{α}^c for every index α . Consequently, x is not an element of E_{α} for any α . Since x is in none of the individual sets, it cannot possibly be in their union. Thus, $x \notin \bigcup_{\alpha} E_{\alpha}$. This directly means x resides in the complement of the union, so $x \in (\bigcup_{\alpha} E_{\alpha})^c$, which means $x \in A$. This establishes that $B \subset A$.

Step 3: Conclusion.

Since we have rigorously shown both $A \subset B$ and $B \subset A$, it logically follows that the sets are strictly equal, so $A = B$. $\square$

Theorem 1.45. A set E is open if and only if its complement is closed.

Proof. Since this is an "if and only if" statement, we must prove the implication in both directions.

Part 1: Assume the complement is closed.

First, let us suppose that the complement E^c is a closed set. Our goal is to prove that E is an open set, which means we must show that every point in E is an interior point.

Choose an arbitrary point $x \in E$. Because x is in E , it is clearly not in the complement, so $x \notin E^c$. We are given that E^c is a closed set, which by definition means it contains all of its limit points. Since x is not contained in E^c , x cannot possibly be a limit point of E^c .

What does it mean for x to *not* be a limit point of E^c ? It means there must exist at least one neighborhood N of x that does not contain any points of E^c . In other words, the intersection $E^c \cap N$ is empty.

If this neighborhood N contains no points from the complement E^c , then every single point inside N must belong to E . Therefore, $N \subset E$. Because we found a neighborhood of x completely contained in E , x is an interior point of E . Since x was chosen arbitrarily, every point in E is an interior point, proving that E is open.

Part 2: Assume the set is open.

Next, let us suppose the converse: E is an open set. We must prove that E^c is a closed set. To do this, we need to show that E^c contains all of its limit points.

Let x be an arbitrary limit point of E^c . By the fundamental definition of a limit point, every neighborhood of x must contain at least one point of E^c .

Because every neighborhood of x contains some point outside of E (since it contains a point of E^c), it is impossible to draw a neighborhood around x that is entirely contained within E . Consequently, x cannot be an interior point of E .

Now we use our initial assumption: E is an open set. This means that every element of E is an interior point. Since we just established that x is not an interior point, it is impossible for x to be an element of E . Thus, $x \notin E$.

If x is not in E , it must belong to its complement, so $x \in E^c$. We have shown that any arbitrary limit point of E^c must belong to E^c . Therefore, E^c contains all of its limit points, proving that E^c is closed. $\square$

Corollary 1.45.1. A set F is closed if and only if its complement F^c is open.

Proof. This follows as an immediate consequence of our previous theorem, utilizing the fundamental fact that the complement of a complement returns the original set (i.e., $(F^c)^c = F$).

Part 1: Assume F is closed.

If F is closed, we can think of F as the complement of some set E (specifically, $E = F^c$). Since F is closed, this means E^c is closed. By our previous theorem, a set E is open if and only if E^c is closed. Therefore, E (which is F^c) must be open.

Part 2: Assume F^c is open.

Conversely, suppose the complement F^c is open. Let us call this open set E (so $E = F^c$). Our previous theorem states that if a set E is open, its complement E^c is closed. Since $E^c = (F^c)^c = F$, it logically follows that F is closed.

Therefore, F is closed if and only if F^c is open. □

Theorem 1.46. *The following properties hold for open and closed sets:*

- (a) *For any collection $\{G_\alpha\}$ of open sets, the union $\bigcup_\alpha G_\alpha$ is open.*
- (b) *For any collection $\{F_\alpha\}$ of closed sets, the intersection $\bigcap_\alpha F_\alpha$ is closed.*
- (c) *For any finite collection $G_1, \dots, G_n$ of open sets, the intersection $\bigcap_{i=1}^n G_i$ is open.*
- (d) *For any finite collection $F_1, \dots, F_n$ of closed sets, the union $\bigcup_{i=1}^n F_i$ is closed.*

Proof. We will prove each part sequentially, using the fundamental definitions of open sets, closed sets, and neighborhoods.

Part (a): Arbitrary union of open sets

Let $G = \bigcup_\alpha G_\alpha$. To prove G is open, we must show every point in G is an interior point. Suppose $x \in G$. By the definition of a union, x must belong to at least one of the sets in the collection, say G_α . Because G_α is given as an open set, x is an interior point of G_α . This means there exists a neighborhood N of x such that $N \subset G_\alpha$. Since G_α is a subset of the entire union G (i.e., $G_\alpha \subset G$), it directly follows that $N \subset G$. Therefore, x is an interior point of G . Since x was arbitrary, G is open.

Part (b): Arbitrary intersection of closed sets

We can prove this elegantly using De Morgan's Laws and our previous knowledge about complements. By De Morgan's Law, the complement of an intersection is the union of the complements:

$$\left(\bigcap_\alpha F_\alpha \right)^c = \bigcup_\alpha (F_\alpha^c)$$

Since each F_α is a closed set, its complement F_α^c must be an open set. The right side of the equation is now an arbitrary union of open sets. By the result we just proved in Part (a), this union must be open. Since the complement of $\bigcap_\alpha F_\alpha$ is open, the set $\bigcap_\alpha F_\alpha$ itself must be closed.

Part (c): Finite intersection of open sets

Let $H = \bigcap_{i=1}^n G_i$. Let $x \in H$. By the definition of intersection, x belongs to every single G_i for $i = 1, \dots, n$. Since each G_i is open, x is an interior point of every G_i . Thus, for each i , there exists a neighborhood N_i of x with some strictly positive radius r_i such that $N_i \subset G_i$. To ensure we find a neighborhood that fits inside *all* G_i simultaneously, we take the smallest of these radii. Let:

$$r = \min(r_1, \dots, r_n)$$

Crucially, because this is a *finite* collection of strictly positive numbers, their minimum r is guaranteed to be strictly positive ($r > 0$). Let N be the neighborhood of x with radius r . Because $r \leq r_i$ for every i , N is contained in every N_i , meaning $N \subset G_i$ for $i = 1, \dots, n$. Therefore, $N \subset H$. This shows x is an interior point of H , and thus H is open.

Part (d): Finite union of closed sets

We apply De Morgan's Laws again, this time for a finite collection. By taking complements, we get:

$$\left(\bigcup_{i=1}^n F_i \right)^c = \bigcap_{i=1}^n (F_i^c)$$

Because each F_i is closed, its complement F_i^c is open. The right side is a finite intersection of open sets, which we just proved in Part (c) must be open. Since the complement of $\bigcup_{i=1}^n F_i$ is open, it follows that the finite union $\bigcup_{i=1}^n F_i$ is closed. □

Example 1.47 (Why "Finite" Matters). In parts (c) and (d) of the previous theorem, we specified that the collections must be finite. This finiteness is absolutely essential, and we can prove why using a simple counterexample.

Let us define a sequence of open intervals (segments) in the real numbers $\mathbb{R}^1$:

$$G_n = \left(-\frac{1}{n}, \frac{1}{n}\right) \quad \text{for } n = 1, 2, 3, \dots$$

Each individual set G_n is clearly an open subset of $\mathbb{R}^1$. As n gets larger (e.g., $n = 1, 2, 10, 100$), the intervals $(-1, 1), (-\frac{1}{2}, \frac{1}{2}), (-\frac{1}{10}, \frac{1}{10})$ shrink tighter and tighter around zero.

Now, let us find the intersection of this entirely infinite collection of sets, which we will call G :

$$G = \bigcap_{n=1}^{\infty} G_n$$

What points survive this infinite intersection? The only number that belongs to every single one of these shrinking intervals is exactly 0. Therefore, the final set G consists of just a single point: $x = 0$.

A set consisting of a single point is not an open subset of $\mathbb{R}^1$ (because you cannot draw a neighborhood with a positive radius around 0 without stepping outside the set $\{0\}$).

This demonstrates that the intersection of an infinite collection of open sets need not be open. By a very similar logic, the union of an infinite collection of closed sets need not be closed.

Definition 1.48. If X is a metric space, if $E \subset X$, and if E' denotes the set of all limit points of E in X , then the closure of E is the set $\overline{E} = E \cup E'$.

Theorem 1.49. If X is a metric space and $E \subset X$, then:

- (a) $\overline{E}$ is closed,
- (b) $E = \overline{E}$ if and only if E is closed,
- (c) $\overline{E} \subset F$ for every closed set $F \subset X$ such that $E \subset F$.

By (a) and (c), $\overline{E}$ is the smallest closed subset of X that contains E .

Proof. Part (a): Let $p \in X$ such that $p \notin \overline{E}$. This means p is neither a point of E nor a limit point of E . Because p is not a limit point, there exists a neighborhood N of p that does not intersect E . Furthermore, N cannot contain any limit points of E (if it did, any neighborhood around that limit point would intersect E , which would force N to intersect E). Therefore, N does not intersect E' either. Thus, $N \subset \overline{E}^c$, meaning every point in the complement of $\overline{E}$ is an interior point. So $\overline{E}^c$ is open, which means $\overline{E}$ is closed.

Part (b): If $E = \overline{E}$, then E is closed because $\overline{E}$ is closed by part (a). Conversely, if E is closed, it contains all of its limit points by definition. Thus, $E' \subset E$, which implies $\overline{E} = E \cup E' = E$.

Part (c): Let F be a closed set such that $E \subset F$. If p is a limit point of E , every neighborhood of p intersects E . Since $E \subset F$, every neighborhood of p also intersects F , making p a limit point of F . Thus, $E' \subset F'$. Because F is closed, it contains all its limit points ($F' \subset F$). Therefore, $E' \subset F$. Since $E \subset F$ and $E' \subset F$, it follows that $\overline{E} = E \cup E' \subset F$. $\square$

Theorem 1.50. Let E be a nonempty set of real numbers which is bounded above. Let $y = \sup E$. Then $y \in \overline{E}$. Hence $y \in E$ if E is closed.

Proof. If $y \in E$, then trivially $y \in \overline{E}$.

Suppose $y \notin E$. By the definition of the supremum, y is the least upper bound of E . For any strictly positive number $h > 0$, the value $y - h$ is strictly less than y , and therefore $y - h$ cannot be an upper bound for E . This means there must exist some point $x \in E$ such that $y - h < x < y$.

Since we can find such an $x \in E$ for any $h > 0$, every neighborhood of y contains a point of E distinct from y . Therefore, y is a limit point of E (i.e., $y \in E'$). Since $y \in E'$, it follows that $y \in \overline{E}$. If E is a closed set, $E = \overline{E}$, and therefore $y \in E$. $\square$

Example 1.51 (Examples of Closure). 1. **Rationals and Irrationals:** Consider the rational numbers $\mathbb{Q}$ and the irrational numbers $\mathbb{R} \setminus \mathbb{Q}$ as subsets of $\mathbb{R}$. Both sets are dense in $\mathbb{R}$. Every real number is a limit point for both sets. Therefore, $\overline{\mathbb{Q}} = \mathbb{R}$ and $\overline{\mathbb{R} \setminus \mathbb{Q}} = \mathbb{R}$.

2. **Relative Closure:** Consider the metric space $X = (0, 2]$ equipped with the standard metric. Let $E = (0, 1)$ be a subset of X . The only limit point of E that actually belongs to the metric space X is the point 1. The point 0 is not in X , so it cannot be a limit point in this space. Therefore, the closure of E in X is $\bar{E} = (0, 1]$.
3. **Sets without Limit Points:** Consider the natural numbers $\mathbb{N}$ as a subset of $\mathbb{R}$. $\mathbb{N}$ has no limit points, so its set of limit points is empty ($\mathbb{N}' = \emptyset$). Therefore, $\bar{\mathbb{N}} = \mathbb{N} \cup \emptyset = \mathbb{N}$. The exact same logic applies to any finite set or any set consisting entirely of isolated points: they contain no limit points, so their closure is simply the set itself.

Definition 1.52. Suppose $E \subset Y \subset X$, where X is a metric space. We say that E is open relative to Y if to each $p \in E$ there is associated an $r > 0$ such that $q \in E$ whenever $d(p, q) < r$ and $q \in Y$. As a consequence, a set may be open relative to Y without being an open subset of the larger space X .

Example 1.53 (Relative Openness in Intervals). Let $X = \mathbb{R}$ with the standard metric, and consider the subspace $Y = (0, 2]$.

First, consider the subset $E = (1, 2]$. In the parent space $X = \mathbb{R}$, E is not open because no standard open interval around the point 2 is entirely contained in E (any interval $(2 - r, 2 + r)$ contains points strictly greater than 2). However, E is open relative to Y . For the point $p = 2 \in E$, we can choose a radius $r = 1$. The relative neighborhood consists of all $q \in Y$ such that $d(2, q) < 1$. These points form the interval $(1, 2]$, which is entirely contained inside E . Thus, $(1, 2]$ is an open set in the relative topology of Y .

Next, consider $E = (0, 1)$. This set is open as a standard subset of $\mathbb{R}$, and it also remains open relative to Y , since restricting its standard neighborhoods to Y keeps them fully contained within E .

Example 1.54 (Relative Openness of Natural Numbers). Consider $Y = \mathbb{N}$ as a subspace of the metric space $X = \mathbb{R}$ with the standard metric. Let $E = \{n\}$ be a singleton set where $n \in \mathbb{N}$. In $\mathbb{R}$, a single point is never an open set.

However, E is open relative to $\mathbb{N}$. If we choose a radius $r = \frac{1}{2}$, the only point $q \in \mathbb{N}$ satisfying $d(n, q) < \frac{1}{2}$ is the point n itself. Thus, the relative neighborhood around n is just $\{n\}$, which is entirely contained in E . Because every singleton is open relative to $\mathbb{N}$, any subset of $\mathbb{N}$ is open relative to $\mathbb{N}$.

Remark 1.55 (Subspace Topology). This concept of relative openness formally describes the subspace topology. When a subset Y inherits its metric from a parent space X , the open sets of Y are exactly the intersections of the open sets of X with Y . As demonstrated in the previous example, the natural numbers $\mathbb{N}$ inherit the discrete topology from the usual topology of $\mathbb{R}$, which fundamentally means that every subset of $\mathbb{N}$ (including singletons) is an open set in that relative space.

Theorem 1.56. Suppose $Y \subset X$. A subset E of Y is open relative to Y if and only if $E = Y \cap G$ for some open subset G of X .

Proof. Because this is an "if and only if" statement, we divide the proof into two parts to show both directions of the implication.

Part 1: Assume E is open relative to Y .

Suppose E is open relative to Y . Our goal is to construct an open set G in the larger space X such that $E = G \cap Y$.

By the definition of relative openness, for every point $p \in E$, there exists a positive radius $r_p > 0$ such that any point q satisfying $d(p, q) < r_p$ and $q \in Y$ must belong to E .

Let us construct a standard open neighborhood in the entire space X around each point $p \in E$ using this specific radius r_p . Let this neighborhood be V_p :

$$V_p = \{q \in X \mid d(p, q) < r_p\}$$

Next, we define our set G as the union of all these individual neighborhoods for every point in E :

$$G = \bigcup_{p \in E} V_p$$

Because every neighborhood V_p is an open set in X , and any arbitrary union of open sets is open, G is guaranteed to be an open subset of X .

Now we must verify that $E = G \cap Y$. First, since every point $p \in E$ is the center of its own neighborhood V_p , it is clear that $p \in G$. Since E is initially given as a subset of Y , it follows that

$E \subset G \cap Y$. Second, by our specific choice of the radius r_p , the portion of any neighborhood V_p that actually lies in Y is completely contained within E . Mathematically, $V_p \cap Y \subset E$ for every $p \in E$. Therefore, the union of all these intersections, which represents $G \cap Y$, must also be a subset of E . So, $G \cap Y \subset E$.

Since $E \subset G \cap Y$ and $G \cap Y \subset E$, we conclude that $E = G \cap Y$.

Part 2: Assume $E = G \cap Y$.

Conversely, suppose there exists some open set G in X such that $E = G \cap Y$. We must prove that E is open relative to Y .

Let p be an arbitrary point in E . Because $E = G \cap Y$, this point p must reside in G . Since G is an open set in X , p is an interior point of G . This means there exists a neighborhood V_p around p (with some positive radius) that is entirely contained within G , meaning $V_p \subset G$.

If we intersect this neighborhood with Y , we get:

$$V_p \cap Y \subset G \cap Y$$

But we are given that $G \cap Y = E$, which means $V_p \cap Y \subset E$.

In practical terms, we have found a radius around p such that all points within that radius that also belong to Y are guaranteed to fall in E . Since this holds for any arbitrary point $p \in E$, the set E is open relative to Y . $\square$

Compact Sets

Definition 1.57. By an open cover of a set E in a metric space X we mean a collection $\{G_\alpha\}$ of open subsets of X such that $E \subset \bigcup_\alpha G_\alpha$.

Definition 1.58. A subset K of a metric space X is said to be compact if every open cover of K contains a finite subcover.

Example 1.59 (Compact and Non-Compact Sets). To understand compactness, it is helpful to look at sets that have this property and sets that fail to have it.

1. Finite Sets (Always Compact)

Any finite set is trivially compact. If a set $K = \{p_1, p_2, \dots, p_n\}$ is covered by an arbitrary open cover $\{G_\alpha\}$, we can simply pick one set from the cover for each point. Let G_1 cover p_1 , G_2 cover p_2 , and so on. The collection $\{G_1, G_2, \dots, G_n\}$ is a finite subcover.

2. The Natural Numbers (Not Compact)

Consider the natural numbers $\mathbb{N}$ as a subset of $\mathbb{R}$ with the standard metric. Let us construct an open cover by taking an open interval of radius 0.1 around each natural number:

$$G_n = (n - 0.1, n + 0.1) \quad \text{for each } n \in \mathbb{N}$$

The collection $\{G_1, G_2, G_3, \dots\}$ is clearly an open cover of $\mathbb{N}$. However, this cover has no finite subcover. Any finite subcollection can only cover a finite number of integers. Because there are infinitely many natural numbers, a finite subcollection will always leave infinitely many points of $\mathbb{N}$ uncovered. Thus, $\mathbb{N}$ is not compact.

3. The Set $E = \{1/n \mid n \in \mathbb{N}\}$ (Not Compact)

Consider the set $E = \{1, 1/2, 1/3, \dots\}$. We can cover this set with the collection of open intervals:

$$U_n = \left(\frac{1}{n}, 2\right) \quad \text{for } n \in \mathbb{N}$$

The union of all U_n covers E . However, if we try to take a finite subcover, it will consist of a finite number of these sets. Let N be the largest index among the sets in our finite subcover. The union of this finite subcover will just be $U_N = (1/N, 2)$. This finite subcover completely fails to cover the points $\frac{1}{N+1}, \frac{1}{N+2}, \dots$ of E . Since we found an open cover that has no finite subcover, E is not compact.

4. The Set $K = \{1/n \mid n \in \mathbb{N}\} \cup \{0\}$ (Compact)

By simply adding the limit point 0 to our previous set, it becomes compact. Let us prove this rigorously.

Let $\{V_\alpha\}$ be an arbitrary open cover of K . Since $0 \in K$, there must be some open set in our cover, let's call it V_0 , that contains the point 0. Because V_0 is an open set, there exists a neighborhood around 0 with some radius $r > 0$ that is entirely contained within V_0 .

By the Archimedean property, we can find a large enough integer N such that $\frac{1}{N} < r$. This means that for all $n \geq N$, the points $\frac{1}{n}$ are strictly inside that neighborhood, and therefore are all safely covered by the single set V_0 .

How many points of K are left uncovered? Only finitely many: $1, 1/2, \dots, \frac{1}{N-1}$. Because there are only finitely many points remaining, we can just pick one open set from our original cover $\{V_\alpha\}$ for each of them. Let's call them $V_1, V_2, \dots, V_{N-1}$.

The finite collection $\{V_0, V_1, V_2, \dots, V_{N-1}\}$ now covers the entire set K . Since we successfully found a finite subcover for an arbitrary open cover, K is compact.

5. The Set $S = \{\frac{1}{n} - \frac{1}{m}\} \cup \{0\}$

Consider the set formed by the differences of unit fractions, together with 0. The limit points of this set include sequences where n is fixed and $m \rightarrow \infty$ (giving $1/n$), sequences where m is fixed and $n \rightarrow \infty$ (giving $-1/m$), and sequences where both go to infinity (giving 0).

Because the set S contains all of these limit points, S is a closed set. Furthermore, because the maximum distance from 0 is bounded by 1, S is also a bounded set. Just like the previous example, this set is compact (any open set covering 0 and the other limit points will cover all but finitely many elements). This dual property—being both closed and bounded—becomes highly relevant very soon when we prove the equivalence of compact sets and closed-and-bounded sets in $\mathbb{R}^k$.

Theorem 1.60. Suppose $K \subset Y \subset X$. Then K is compact relative to X if and only if K is compact relative to Y .

Proof. We need to prove this equivalence in both directions.

Part 1: Assume K is compact relative to X .

Let $\{V_\alpha\}$ be an arbitrary collection of sets, open relative to Y , that covers K (so $K \subset \bigcup_\alpha V_\alpha$). By our previous theorem on subspace topologies, for each set V_α that is open relative to Y , there exists a corresponding set G_α that is open in the larger space X such that $V_\alpha = Y \cap G_\alpha$.

Because the sets V_α cover K , and each V_α is a subset of G_α , the collection of open sets $\{G_\alpha\}$ clearly covers K in X . Since we assumed K is compact relative to X , this open cover $\{G_\alpha\}$ must have a finite subcover. Thus, there exist finitely many indices $\alpha_1, \dots, \alpha_n$ such that:

$$K \subset G_{\alpha_1} \cup \dots \cup G_{\alpha_n}$$

Because K is entirely contained within Y , intersecting the right side of the above inclusion with Y will still fully cover K . Therefore:

$$K \subset (Y \cap G_{\alpha_1}) \cup \dots \cup (Y \cap G_{\alpha_n})$$

Substituting back our definition of V_α , we get:

$$K \subset V_{\alpha_1} \cup \dots \cup V_{\alpha_n}$$

We have successfully found a finite subcover from our original relative open cover $\{V_\alpha\}$. This proves that K is compact relative to Y .

Part 2: Assume K is compact relative to Y .

Conversely, suppose $\{G_\alpha\}$ is a collection of open subsets in X that covers K . For each index α , we define $V_\alpha = Y \cap G_\alpha$. Because each G_α is open in X , each V_α is open relative to Y .

Since $K \subset Y$ and K is covered by the sets G_α , it follows that K is covered by their intersections with Y . Thus, $\{V_\alpha\}$ is an open cover of K relative to Y . By our assumption that K is compact relative to Y , there must exist a finite subcover. Therefore, for some finitely many indices $\alpha_1, \dots, \alpha_n$:

$$K \subset V_{\alpha_1} \cup \dots \cup V_{\alpha_n}$$

Since $V_\alpha \subset G_\alpha$ for every index, it immediately follows that:

$$K \subset G_{\alpha_1} \cup \dots \cup G_{\alpha_n}$$

We have found a finite subcover from the original open cover $\{G_\alpha\}$ in X . This completes the proof that K is compact relative to X . $\square$

Remark 1.61 (Absolute Nature of Compactness). By virtue of this theorem, we can regard compact sets as metric spaces in their own right, without paying attention to the embedding space. Openness and closedness are purely relative properties (they depend heavily on the parent space), but compactness is an absolute property. Proving a subset is compact within a subspace under the inherited metric is mathematically identical to proving it is compact in the whole larger metric space.

Example 1.62. Consider the metric space $X = \mathbb{R}$ with the standard metric, and the subspace $Y = (0, 2]$. By our theorem, any subset of $(0, 2]$ that is compact relative to $(0, 2]$ will also be compact in $\mathbb{R}$, and vice versa. However, this stability absolutely does not hold for closed and open sets:

- The set $(0, 2]$ is **closed** in $(0, 2]$ (because every metric space is a closed subset of itself), but it is **not closed** in $\mathbb{R}$ (because it fails to contain its limit point 0).
- The set $(1, 2]$ is **open** in $(0, 2]$ (it is the intersection of the standard open interval $(1, 3)$ with Y), but it is **not open** in $\mathbb{R}$ (because no standard open interval centered at 2 is entirely contained within the set).

Compactness transcends this relativity, meaning a compact set remains intrinsically compact regardless of what larger space you place it in.

Theorem 1.63. Compact subsets of metric spaces are closed.

Proof. Let K be a compact subset of a metric space X . To prove K is closed, we will establish that its complement, K^c , is an open set.

Suppose p is an arbitrary point in the complement K^c (meaning $p \notin K$). Our objective is to find a neighborhood around p that is entirely contained within K^c .

For every point $q \in K$, the distance $d(p, q)$ is strictly positive because p and q are distinct points. Let us construct two non-overlapping neighborhoods for each q :

1. A neighborhood V_q centered at p with a radius less than $\frac{1}{2}d(p, q)$.
2. A neighborhood W_q centered at q with a radius less than $\frac{1}{2}d(p, q)$.

Because the radii are less than half the distance between p and q , these two neighborhoods are guaranteed to be disjoint.

Now, consider the collection of all these neighborhoods $\{W_q\}$ for every $q \in K$. This collection forms an open cover of the set K . Since K is given to be compact, this arbitrary open cover must contain a finite subcover. Therefore, there are finitely many points $q_1, \dots, q_n$ in K such that their corresponding neighborhoods completely cover K :

$$K \subset W_{q_1} \cup \dots \cup W_{q_n} = W$$

For each of these specifically chosen points q_i , there is a corresponding neighborhood V_{q_i} centered around our original point p . Let V be the intersection of these finitely many neighborhoods around p :

$$V = V_{q_1} \cap \dots \cap V_{q_n}$$

Since it is an intersection of a finite number of open neighborhoods, V itself is an open neighborhood of p .

Because each individual V_{q_i} is completely disjoint from its corresponding W_{q_i} , their intersection V is completely disjoint from the entire union W . Since K is a subset of W , V does not intersect K at all. This means $V \subset K^c$.

We have successfully found a neighborhood V around an arbitrary point $p \in K^c$ that lies entirely outside of K . Therefore, p is an interior point of K^c , making K^c an open set. Consequently, K is closed. $\square$

Theorem 1.64. Closed subsets of compact sets are compact.

Proof. Suppose $F \subset K \subset X$, where F is a closed set (relative to the metric space X) and K is a compact set. We must prove that F is compact.

Let $\{V_\alpha\}$ be an arbitrary open cover of F . To prove compactness, we need to extract a finite subcover from it.

Since F is a closed set, its complement F^c is an open set. Let us add this single open set F^c to our covering collection. We will define a new collection of open sets Ω :

$$\Omega = \{V_\alpha\} \cup \{F^c\}$$

This new collection Ω is an open cover for the entire set K . To see why, pick any point in K . It either belongs to F (in which case it is covered by some V_α) or it does not belong to F (in which case it is covered by F^c).

Since K is a compact set, every open cover of K must have a finite subcover. So there exists a finite subcollection Φ chosen from Ω that completely covers K . Because $F \subset K$, this finite subcollection Φ also completely covers F .

This finite subcollection Φ might happen to include the set F^c . If it does, we simply discard F^c from Φ . Because F^c contains absolutely no points of F , removing it from our covering collection does not leave any points of F exposed.

The remaining sets in our finite subcollection belong entirely to the original cover $\{V_\alpha\}$ and still completely cover F . We have successfully found a finite subcover for F , which proves that F is compact. $\square$

Corollary 1.64.1. *If F is closed and K is compact, then $F \cap K$ is compact.*

Proof. By our above theorem, any compact subset of a metric space is closed. Since K is given as a compact set, K must be closed. We are also given that F is a closed set. A fundamental property of open sets dictates that the intersection of any collection of closed sets is closed. Therefore, the intersection $F \cap K$ is a closed set.

Furthermore, it is trivially true that $F \cap K \subset K$. We have just established that $F \cap K$ is a closed subset of the compact set K . By our second theorem (which states that closed subsets of compact sets are compact), it directly follows that $F \cap K$ is compact. $\square$

Theorem 1.65. *If $\{K_\alpha\}$ is a collection of compact subsets of a metric space X such that the intersection of every finite subcollection of $\{K_\alpha\}$ is nonempty, then the total intersection $\bigcap K_\alpha$ is nonempty.*

Proof. We will prove this by contradiction.

Step 1: Set up the contradiction.

Fix one specific compact set from the collection and call it K_1 . Because each K_α is a compact subset of a metric space, it must be closed. Therefore, its complement $G_\alpha = K_\alpha^c$ is an open set.

Now, assume for the sake of contradiction that the total intersection $\bigcap K_\alpha$ is empty. This means there is absolutely no point that belongs to *every* set K_α . In particular, no point in our fixed set K_1 belongs to every K_α .

Step 2: Construct an open cover.

If no point of K_1 is in every K_α , then every point of K_1 is missing from at least one K_α . If a point is missing from K_α , it must belong to its complement, G_α . This implies that the collection of open sets $\{G_\alpha\}$ completely covers K_1 . Thus, $\{G_\alpha\}$ forms an open cover of K_1 .

Step 3: Use compactness to find a finite subcover.

We are given that K_1 is a compact set. By the fundamental definition of compactness, this open cover must have a finite subcover. Thus, there exist finitely many indices $\alpha_1, \dots, \alpha_n$ such that:

$$K_1 \subset G_{\alpha_1} \cup \dots \cup G_{\alpha_n}$$

Step 4: Reach a contradiction.

Let us translate this back to our original sets. The union of the complements G_{α_i} is exactly the complement of the intersection of the sets K_{α_i} . If K_1 is entirely contained within the complement of $(K_{\alpha_1} \cap \dots \cap K_{\alpha_n})$, it means K_1 shares absolutely no elements with that intersection.

Mathematically, this means:

$$K_1 \cap K_{\alpha_1} \cap \dots \cap K_{\alpha_n} = \emptyset$$

However, this is a finite subcollection of our original sets, and our initial hypothesis explicitly stated that *every* finite subcollection has a nonempty intersection. We have reached a direct contradiction.

Therefore, our assumption must be false, and the total intersection $\bigcap K_\alpha$ cannot be empty. $\square$

Corollary 1.65.1. *If $\{K_n\}$ is a sequence of nonempty compact sets such that $K_n \supset K_{n+1}$ (for $n = 1, 2, 3, \dots$), then $\bigcap_{n=1}^{\infty} K_n$ is not empty.*

Proof. This follows directly as a special case of the preceding theorem.

Consider any arbitrary finite subcollection of these sets, say $K_{n_1}, K_{n_2}, \dots, K_{n_m}$. Let N be the largest index among them ($N = \max(n_1, \dots, n_m)$).

Because the sequence of sets is nested ($K_1 \supset K_2 \supset K_3 \dots \supset K_N$), the set K_N with the largest index is the smallest set in the chain and is completely contained within all the earlier sets in this subcollection. Therefore, the intersection of this entire finite subcollection is simply equal to K_N .

Since we are given that every individual set in the sequence is nonempty, K_N is nonempty. We have just shown that the intersection of *any* finite subcollection is nonempty. By the previous theorem, it immediately follows that the total intersection of all the sets, $\bigcap_{n=1}^{\infty} K_n$, is not empty. $\square$

Theorem 1.66. *If E is an infinite subset of a compact set K , then E has a limit point in K .*

Proof. We will prove this by contradiction. Assume that no point of K is a limit point of E .

By the definition of a limit point, if a point $q \in K$ is not a limit point of E , there must exist a neighborhood V_q around q that contains no points of E other than possibly q itself. Thus, each neighborhood V_q contains at most one point of E (specifically, it contains exactly q if $q \in E$, and zero points of E if $q \notin E$).

Now, construct a collection of these neighborhoods $\{V_q\}$ for every single point $q \in K$. Because each point q is the center of its own neighborhood, this collection clearly covers all of K . Therefore, $\{V_q\}$ is an open cover of K .

Since we are given that K is a compact set, this open cover must contain a finite subcover. That means there is a finite number of neighborhoods, say $V_{q_1}, V_{q_2}, \dots, V_{q_n}$, that completely cover K .

Because E is a subset of K , this finite subcover must also cover E . However, we established that each individual neighborhood V_{q_i} contains at most one point of E . Therefore, the union of these finitely many neighborhoods can contain, at most, n points of E .

This implies that the set E has only finitely many points. But our initial premise explicitly stated that E is an infinite subset. We have reached a direct contradiction.

Therefore, our assumption must be false, and E must possess at least one limit point in K . $\square$

Theorem 1.67. *If $\{I_n\}$ is a sequence of intervals in $\mathbb{R}^1$, such that $I_n \supset I_{n+1}$ for $n = 1, 2, 3, \dots$, then the intersection $\bigcap_{n=1}^{\infty} I_n$ is not empty.*

Proof. Let each interval be denoted by its endpoints: $I_n = [a_n, b_n]$. The nested condition $I_n \supset I_{n+1}$ implies that for every n , the next interval is contained within the previous one, which means $a_n \leq a_{n+1}$ and $b_{n+1} \leq b_n$.

Let E be the set of all the left endpoints: $E = \{a_1, a_2, a_3, \dots\}$. This set E is clearly nonempty. Furthermore, because every interval I_n is contained within the very first interval $I_1 = [a_1, b_1]$, every left endpoint a_n must be less than or equal to b_1 . Thus, E is bounded above by b_1 .

By the Least Upper Bound Property of the real numbers, any nonempty set that is bounded above has a supremum. Let $x = \sup E$.

We want to show that this specific point x belongs to every interval I_m (which would prove the total intersection is not empty). To do this, we need to show that $a_m \leq x \leq b_m$ for any positive integer m .

First, because x is the supremum (the least upper bound) of the set of all left endpoints E , it must be greater than or equal to every element in E . Therefore, it is obvious that $a_m \leq x$ for all m .

Second, let us take any two positive integers m and n . Because the intervals are nested, we can relate their endpoints:

$$a_n \leq a_{m+n} \leq b_{m+n} \leq b_m$$

This inequality shows that *every* right endpoint b_m is an upper bound for the entire set of left endpoints E . Since x is the *least* upper bound of E , it must be less than or equal to any other upper bound. Therefore, $x \leq b_m$ for each m .

Since we have shown that $a_m \leq x \leq b_m$ for every index $m = 1, 2, 3, \dots$, it follows that $x \in I_m$ for every m . Thus, $x \in \bigcap_{n=1}^{\infty} I_n$, proving the intersection is not empty. $\square$

Theorem 1.68. *Let k be a positive integer. If $\{I_n\}$ is a sequence of k -cells such that $I_n \supset I_{n+1}$ ($n = 1, 2, 3, \dots$), then $\bigcap_{n=1}^{\infty} I_n$ is not empty.*

Proof. A k -cell in $\mathbb{R}^k$ can be viewed as the Cartesian product of k closed, one-dimensional intervals. Let us break down each k -cell I_n into its constituent coordinate intervals.

Let I_n consist of all points $\mathbf{x} = (x_1, \dots, x_k)$ such that for each coordinate index j ($1 \leq j \leq k$), the coordinate is bounded by:

$$a_{n,j} \leq x_j \leq b_{n,j} \quad (n = 1, 2, 3, \dots)$$

We can denote the closed interval for the j -th coordinate of the n -th cell as $I_{n,j} = [a_{n,j}, b_{n,j}]$.

We are given that the k -cells are nested, meaning $I_n \supset I_{n+1}$ for every n . For a larger k -cell to completely contain a smaller one, every single 1-dimensional coordinate interval of the larger cell must contain the corresponding coordinate interval of the smaller cell. Therefore, for each fixed coordinate dimension j , we have a nested sequence of 1-dimensional intervals:

$$I_{n,j} \supset I_{n+1,j} \quad \text{for all } n$$

Now, we can look at each of the k dimensions independently. For any fixed j , the sequence $\{I_{n,j}\}$ is a sequence of nested closed intervals in $\mathbb{R}^1$. By the above theorem (regarding nested sequences of intervals in $\mathbb{R}^1$), we know that the intersection of such a sequence is not empty.

Consequently, for each coordinate index $j \in \{1, 2, \dots, k\}$, there must exist at least one real number, let's call it x_j^* , that belongs to the intersection of all the intervals in that dimension. Mathematically, this means:

$$a_{n,j} \leq x_j^* \leq b_{n,j} \quad \text{for all } n = 1, 2, 3, \dots$$

Now we assemble these k individual coordinates back into a single point in $\mathbb{R}^k$. Let us construct the point $\mathbf{x}^* = (x_1^*, x_2^*, \dots, x_k^*)$.

Because every single coordinate x_j^* of this point satisfies the bounding inequalities defining the k -cell I_n , the point $\mathbf{x}^*$ must belong to I_n . Furthermore, because these inequalities hold for *all* $n \geq 1$, the point $\mathbf{x}^*$ belongs to *every* k -cell in the sequence.

Thus, $\mathbf{x}^* \in \bigcap_{n=1}^{\infty} I_n$. We have found a point that exists in the total intersection, completely proving that the intersection is not empty. $\square$

Theorem 1.69. *Every k -cell is compact.*

Proof. We will prove this step-by-step using a method of contradiction and repeated subdivision.

Step 1: The Core Idea and Initial Setup

Let I be a k -cell. By definition, I consists of all points $\mathbf{x} = (x_1, \dots, x_k)$ such that $a_j \leq x_j \leq b_j$ for each dimension j ($1 \leq j \leq k$).

The maximum possible distance between any two points in this k -cell occurs between opposite corners (the main diagonal). We will denote this maximum distance as δ . Using the standard Euclidean distance formula, we calculate this diagonal as:

$$\delta = \left(\sum_{j=1}^k (b_j - a_j)^2 \right)^{1/2}$$

Thus, for any two points $\mathbf{x}, \mathbf{y} \in I$, the distance between them is bounded: $|\mathbf{x} - \mathbf{y}| \leq \delta$.

To proceed by contradiction, assume the contrary: suppose there exists an open cover $\{G_\alpha\}$ of the k -cell I that contains *no finite subcover*.

Step 2: Constructing a Sequence of Nested k -cells

We will now systematically divide our k -cell. For each coordinate interval $[a_j, b_j]$, find its exact midpoint $c_j = (a_j + b_j)/2$. By cutting every one of the k dimensions in half, we divide our original k -cell I into 2^k smaller, equal-sized k -cells whose union is the original cell I .

Since we assumed the entire cell I cannot be covered by any finite subcollection of $\{G_\alpha\}$, it is a logical necessity that at least one of these 2^k smaller sub-cells also cannot be finitely covered. (If all 2^k sub-cells could be finitely covered, we could just combine those finite covers to finitely cover the whole cell I , which contradicts our assumption).

Pick one of these smaller sub-cells that cannot be finitely covered, and call it I_1 . We then repeat this exact subdivision process on I_1 , slicing it into 2^k even smaller cells, at least one of which cannot be finitely covered. Call that one I_2 .

Step 3: Properties of the Sequence

By continuing this process infinitely, we obtain a sequence of k -cells $\{I_n\}$ that possesses three critical properties:

- (a) **Nested:** Each cell is completely contained within the previous one: $I \supset I_1 \supset I_2 \supset I_3 \supset \dots$
- (b) **Uncoverable:** For every integer n , the cell I_n cannot be covered by any finite subcollection of $\{G_\alpha\}$.
- (c) **Shrinking Diagonal:** If $\mathbf{x} \in I_n$ and $\mathbf{y} \in I_n$, then $|\mathbf{x} - \mathbf{y}| \leq 2^{-n}\delta$. (Why the 2^n denominator? Each time we subdivide a cell, we cut the length of every side exactly in half. After n subdivisions, the length of every side is divided by 2^n . By the Euclidean distance formula, if every side shrinks by a factor of 2^n , the overall diagonal maximum distance also shrinks exactly by that same factor of 2^n).

Step 4: Finding the Common Point

By property (a), we have a nested sequence of k -cells. By the above theorem (regarding nested sequences of k -cells), there must exist at least one point, let us call it $\mathbf{x}^*$, that lies in *every* single cell I_n in the sequence.

Step 5: Using the Openness of the Cover

Because the original collection $\{G_\alpha\}$ covers the entire initial cell I , and $\mathbf{x}^* \in I$, this point $\mathbf{x}^*$ must belong to at least one open set in the cover. Let us call this set G_α .

Because G_α is an open set, $\mathbf{x}^*$ is an interior point. This means we can draw a neighborhood around $\mathbf{x}^*$ with some strictly positive radius $r > 0$ that is entirely contained within G_α . In other words, any point $\mathbf{y}$ satisfying $|\mathbf{y} - \mathbf{x}^*| < r$ is guaranteed to be inside G_α .

Step 6: Reaching the Contradiction

We want to show that for a large enough n , the entire cell I_n is swallowed completely by this neighborhood. The maximum size (diagonal) of the cell I_n is $\delta/2^n$. We need this maximum size to be strictly less than our radius r , which means we need $\delta/2^n < r$, or equivalently, $2^n > \delta/r$.

Because the sequence 2^n grows infinitely large as n increases, the Archimedean property of the real numbers guarantees that we can find an integer n large enough such that 2^n surpasses the fixed number δ/r .

Let us pick such a large n . For this specific cell I_n , its maximum diagonal $\delta/2^n$ is strictly less than r . Now, take any arbitrary point $\mathbf{y} \in I_n$. Because both $\mathbf{y}$ and $\mathbf{x}^*$ belong to I_n , the distance between them is at most the diagonal of the cell:

$$|\mathbf{y} - \mathbf{x}^*| \leq 2^{-n}\delta < r$$

Because the distance from $\mathbf{y}$ to $\mathbf{x}^*$ is less than r , every single point $\mathbf{y}$ in the cell I_n falls perfectly inside our neighborhood, and therefore falls inside the single open set G_α . This means $I_n \subset G_\alpha$.

We have just shown that the entire cell I_n can be covered by exactly *one* open set from our cover. However, property (b) explicitly stated that I_n could not be covered by *any* finite subcollection of $\{G_\alpha\}$. One set is a finite subcollection. This is a direct contradiction.

Therefore, our initial assumption must be false. Every open cover of a k -cell must have a finite subcover, making every k -cell compact. $\square$

Theorem 1.70. *If a set E in $\mathbb{R}^k$ has one of the following three properties, then it has the other two:*

- (a) E is closed and bounded.
- (b) E is compact.
- (c) Every infinite subset of E has a limit point in E .

Proof. We will prove this by establishing a chain of implications: (a) implies (b), (b) implies (c), and finally (c) implies (a). This will show that all three statements are logically equivalent.

Step 1: Prove that (a) implies (b)

Assume that statement (a) holds, meaning E is both closed and bounded. Because E is bounded, it can be entirely enclosed within some sufficiently large k -cell, which we will call I . Thus, $E \subset I$. By the above theorem (regarding k -cells), every k -cell is compact. Therefore, I is a compact set. We now know that E is a closed subset that is contained within the compact set I . By the above theorem (regarding subsets), closed subsets of compact sets are compact. Therefore, it immediately follows that E is compact. This proves (b).

Step 2: Prove that (b) implies (c)

Assume that statement (b) holds, meaning E is compact. Let S be any arbitrary infinite subset of E . By the above theorem (regarding infinite subsets of compact sets), if a set is an infinite subset of a compact set, then it must possess a limit point within that compact set. Since S is an infinite subset of the compact set E , it directly follows from this previous theorem that S has a limit point in E . This proves (c).

Step 3: Prove that (c) implies (a)

Assume that statement (c) holds, meaning every infinite subset of E has a limit point in E . We must prove that E is both bounded and closed. We will do this in two sub-parts using proof by contradiction.

Sub-part 3.1: Proving E is bounded.

Suppose, for the sake of contradiction, that E is not bounded. This means we can find points in E that are arbitrarily far from the origin. For each positive integer $n = 1, 2, 3, \dots$, we can select a point $\mathbf{x}_n \in E$ such that its distance from the origin is strictly greater than n ($|\mathbf{x}_n| > n$). Let S be the set consisting of all these chosen points $\mathbf{x}_n$. Because the distances keep growing without bound, all these points are

distinct, making S an infinite subset of E . However, the set S clearly has no limit point anywhere in $\mathbb{R}^k$. For any proposed limit point, a bounded neighborhood around it could only ever contain at most finitely many points of S , because the points of S eventually escape and stay outside any bounded region. Because S has no limit point in $\mathbb{R}^k$, it certainly has no limit point in E . This is an infinite subset of E with no limit point in E , which directly contradicts our assumption (c). Therefore, E must be bounded.

Sub-part 3.2: Proving E is closed.

Suppose, for the sake of contradiction, that E is not closed. By definition, this means there exists some point $\mathbf{x}_0 \in \mathbb{R}^k$ which is a limit point of E , but $\mathbf{x}_0$ is not an element of E ($\mathbf{x}_0 \notin E$). Because $\mathbf{x}_0$ is a limit point of E , every single neighborhood of $\mathbf{x}_0$ must contain infinitely many points of E . We will now construct a specific sequence of points. For each integer $n = 1, 2, 3, \dots$, consider a neighborhood around $\mathbf{x}_0$ of radius $1/n$. Because this is a neighborhood of a limit point, we can pick a point $\mathbf{x}_n \in E$ such that the distance between them is strictly less than this radius:

$$|\mathbf{x}_n - \mathbf{x}_0| < \frac{1}{n}$$

Let S be the set of these specifically chosen points $\mathbf{x}_n$. The set S must be infinite. (If S were finite, the distances $|\mathbf{x}_n - \mathbf{x}_0|$ would reach some minimum positive value and could not keep getting smaller than $1/n$ for arbitrarily large n). By construction, the points of S get arbitrarily close to $\mathbf{x}_0$, making $\mathbf{x}_0$ a limit point of S . However, we already know that $\mathbf{x}_0 \notin E$.

To complete the contradiction with (c), we must show that S has *no other* limit point in E . We will prove it has no other limit point in the entirety of $\mathbb{R}^k$. Let $\mathbf{y}$ be any arbitrary point in $\mathbb{R}^k$ such that $\mathbf{y} \neq \mathbf{x}_0$. Let the distance between these two distinct points be $d = |\mathbf{x}_0 - \mathbf{y}|$. Since they are distinct, $d > 0$. We will choose a radius exactly equal to half the distance between $\mathbf{x}_0$ and $\mathbf{y}$: let $r = d/2$. We want to examine the neighborhood around $\mathbf{y}$ with this radius r . By the Reverse Triangle Inequality, the distance from our arbitrary point $\mathbf{y}$ to any sequence point $\mathbf{x}_n$ is bounded below:

$$|\mathbf{x}_n - \mathbf{y}| \geq |\mathbf{x}_0 - \mathbf{y}| - |\mathbf{x}_n - \mathbf{x}_0|$$

We know $|\mathbf{x}_0 - \mathbf{y}| = d$. We also know that for all points in our sequence, $|\mathbf{x}_n - \mathbf{x}_0| < 1/n$. Therefore:

$$|\mathbf{x}_n - \mathbf{y}| > d - \frac{1}{n}$$

Because $1/n$ shrinks toward zero as n increases, there will be some stage in the sequence (some sufficiently large integer N) after which $1/n \leq d/2$. For all terms in the sequence after this stage ($n \geq N$), we have:

$$|\mathbf{x}_n - \mathbf{y}| > d - \frac{d}{2} = \frac{d}{2} = r$$

This strictly shows that after a certain stage N , every single point $\mathbf{x}_n$ in our sequence is at a distance strictly greater than r from $\mathbf{y}$. Consequently, the neighborhood of $\mathbf{y}$ with radius $r = d/2$ contains absolutely no elements of the sequence S after stage N . Thus, this neighborhood contains at most finitely many points of S (specifically, only those from before stage N). By the above theorem (regarding limit points), every neighborhood of a true limit point *must* contain infinitely many points of the set. Since we found a neighborhood of $\mathbf{y}$ that contains only finitely many points of S , the point $\mathbf{y}$ cannot be a limit point of S .

Because $\mathbf{y}$ was chosen as any arbitrary point other than $\mathbf{x}_0$, we conclude that S has no limit points other than $\mathbf{x}_0$. Since $\mathbf{x}_0 \notin E$, the infinite subset S has no limit point in E . This is a direct contradiction of our assumption (c). Therefore, our premise was false, and E must be closed. Since we have shown (c) implies E is bounded and (c) implies E is closed, we have shown (c) implies (a). The proof of equivalence is complete. $\square$

Remark 1.71. *The assumption that the set E lies in $\mathbb{R}^k$ is crucial for the theorem above. In a general metric space, it is not always true that a closed and bounded set is compact.*

- **Counterexample in a general metric space:** Consider the metric space $X = (0, 2]$ equipped with the standard absolute value metric. Let $E = (0, 1]$. Clearly, E is bounded. Furthermore, E is closed relative to X , because its complement in X is $(1, 2]$, which is an open set in X . However, E is not compact. We know that compactness is an absolute, intrinsic property that does not change depending on the surrounding metric space. If we view $E = (0, 1]$ as a subset of $\mathbb{R}$, it is not closed (it is missing the limit point 0) and thus cannot be compact. Because it is not compact in $\mathbb{R}$, it cannot be compact in X . This demonstrates that being "closed and bounded" is not sufficient to guarantee compactness in arbitrary metric spaces.

- **The general metric space equivalent:** For general metric spaces, the appropriate analog to this theorem involves completeness. A subset of a general metric space is compact if and only if it is complete (meaning every Cauchy sequence converges) and totally bounded.
- **Compact implies closed (Topological perspective):** While "closed and bounded implies compact" fails in general metric spaces, the reverse implication—that every compact set is closed—is true in every metric space. From a broader topological perspective, this is because every metric space is a **Hausdorff space**. If you closely observe the proof that compact sets are closed, it fundamentally relies on the ability to take two distinct points and separate them using disjoint neighborhoods (choosing a radius equal to half the distance between them). This separation property is exactly what defines a Hausdorff space. Therefore, the rule "compact implies closed" is not universally true for all abstract topological spaces, but it is always true in Hausdorff spaces (and consequently, all metric spaces).

Theorem 1.72 (Weierstrass). Every bounded infinite subset of $\mathbb{R}^k$ has a limit point in $\mathbb{R}^k$.

Proof. Let E be any bounded and infinite subset of $\mathbb{R}^k$.

Because the set E is bounded, its spatial extent is limited in every direction. This guarantees that we can construct a sufficiently large k -cell, which we will denote as I , that completely encloses the entirety of E . Therefore, we have $E \subset I \subset \mathbb{R}^k$.

By the above theorem (regarding the compactness of k -cells), we established that every k -cell is a compact set. Consequently, the enclosing k -cell I is compact.

At this stage, we have framed E as an infinite subset that is contained entirely within the compact set I .

By the above theorem (regarding infinite subsets of compact sets), we know that if a set is an infinite subset of a compact set, it must possess a limit point that belongs to that compact set.

Applying this directly, because E is an infinite subset of the compact k -cell I , the set E must have a limit point that lies within I . Since I is just a region within the broader space $\mathbb{R}^k$, this limit point inherently exists in $\mathbb{R}^k$.

Thus, every bounded infinite subset of $\mathbb{R}^k$ has a limit point in $\mathbb{R}^k$. $\square$

Perfect Sets

Theorem 1.73. Let P be a nonempty perfect set in $\mathbb{R}^k$. Then P is uncountable.

Proof. By definition, a set P is *perfect* if it is closed and every point of P is a limit point of P .

Because P is nonempty and every point in P is a limit point, P must contain infinitely many points. We will prove that P is uncountable by contradiction.

Step 1: Set up the proof by contradiction.

Suppose P is countable. Then we can arrange all the elements of P into a sequence and enumerate them as:

$$P = \{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \dots\}$$

We will now construct a nested sequence of neighborhoods $\{V_n\}$ to derive a contradiction.

Step 2: Construct the sequence of neighborhoods $\{V_n\}$.

Let V_1 be any open neighborhood around the first point $\mathbf{x}_1$. The closure of V_1 , denoted by $\overline{V}_1$, is closed and bounded.

Now, suppose we have constructed an open neighborhood V_n such that $V_n \cap P$ is nonempty. We wish to construct the next neighborhood V_{n+1} .

- Since $V_n \cap P$ is nonempty, pick a point $\mathbf{y} \in V_n \cap P$.
- Because every point of P is a limit point of P , the open neighborhood V_n must contain infinitely many points of P .
- Therefore, we can choose a point $\mathbf{y} \in V_n \cap P$ such that $\mathbf{y} \neq \mathbf{x}_n$.
- Since V_n is an open set and $\mathbf{y} \neq \mathbf{x}_n$, we can draw a sufficiently small neighborhood V_{n+1} around $\mathbf{y}$ such that its closure $\overline{V}_{n+1}$ is small enough to satisfy three conditions:

- (i) $\overline{V}_{n+1} \subset V_n$

- (ii) $\mathbf{x}_n \notin \overline{V}_{n+1}$
- (iii) $V_{n+1} \cap P \neq \emptyset$ (since $\mathbf{y} \in V_{n+1} \cap P$)

By condition (iii), V_{n+1} satisfies our inductive hypothesis, and we can continue this process indefinitely for all $n = 1, 2, 3, \dots$.

Step 3: Define a sequence of compact sets K_n .

For each $n \geq 1$, define the set:

$$K_n = \overline{V}_n \cap P$$

Let us analyze the properties of $\{K_n\}$:

1. **Compactness:** The closure $\overline{V}_n$ is closed and bounded in $\mathbb{R}^k$, so it is compact. Since P is a perfect set, P is closed. Thus, K_n is the intersection of a compact set and a closed set, making K_n a closed subset of a compact set. By the above theorem (regarding closed subsets of compact sets), each K_n is compact.
2. **Nonempty:** By condition (iii) above, $V_n \cap P$ is nonempty. Since $V_n \subset \overline{V}_n$, it follows that $K_n = \overline{V}_n \cap P$ is nonempty for every n .
3. **Nested:** By condition (i), $\overline{V}_{n+1} \subset V_n \subset \overline{V}_n$, which implies $K_{n+1} \subset K_n$ for all n .

Step 4: Reach a contradiction.

We have built a sequence $\{K_n\}$ of nonempty compact sets such that $K_1 \supset K_2 \supset K_3 \dots$.

By the above theorem (regarding nested sequences of nonempty compact sets), the total intersection of this sequence must be nonempty:

$$\bigcap_{n=1}^{\infty} K_n \neq \emptyset$$

On the other hand, consider the points of P . By condition (ii) of our construction, $\mathbf{x}_n \notin \overline{V}_{n+1}$. Consequently, $\mathbf{x}_n \notin K_{n+1}$, which means $\mathbf{x}_n$ cannot belong to the total intersection $\bigcap_{i=1}^{\infty} K_i$.

Since every point $\mathbf{x}_n \in P$ is excluded from the intersection, **no point of P lies in $\bigcap_{n=1}^{\infty} K_n$ **.

However, by definition, $K_n = \overline{V}_n \cap P \subset P$ for every n . Therefore, the total intersection $\bigcap_{n=1}^{\infty} K_n$ must be a subset of P . If no point of P lies inside this intersection, then the intersection must be empty:

$$\bigcap_{n=1}^{\infty} K_n = \emptyset$$

This directly contradicts our earlier result that the intersection is nonempty.

Thus, our initial assumption that P is countable must be false. We conclude that P is uncountable. $\square$

Remark 1.74. The assumption that the space is $\mathbb{R}^k$ (or more generally, a complete metric space) is vital here. In an arbitrary metric space, a nonempty perfect set is not guaranteed to be uncountable.

- **Counterexample in $\mathbb{Q}$:** Consider the set of rational numbers $\mathbb{Q}$ as a metric space in its own right, equipped with the standard metric $d(x, y) = |x - y|$. Now consider the subset $P = \mathbb{Q}$ within this metric space $\mathbb{Q}$.

- P is closed in $\mathbb{Q}$ (since any space is closed relative to itself).
- Every point of $\mathbb{Q}$ is a limit point of $\mathbb{Q}$, because for any rational number, we can construct a sequence of distinct rational numbers that converges to it.

Therefore, $P = \mathbb{Q}$ is a perfect set in the metric space $(\mathbb{Q}, d)$. However, $\mathbb{Q}$ is countable! (The proof fails here because closed and bounded neighborhood closures in $\mathbb{Q}$ are not compact, so the nested compact sets theorem cannot be applied.)

- Note that when $\mathbb{Q}$ is viewed as a subset of $\mathbb{R}$ with the standard metric, it is not closed (since its limit points include all irrational numbers), and hence it is not perfect in $\mathbb{R}$.

Corollary 1.74.1. Every interval $[a, b]$ with $a < b$ is uncountable. In particular, the set of all real numbers $\mathbb{R}$ is uncountable.

Proof. Let $I = [a, b]$ with $a < b$ be a closed interval in $\mathbb{R}^1$.

Step 1: Show $[a, b]$ is a perfect set.

1. The interval $[a, b]$ is closed in $\mathbb{R}^1$.
2. Take any point $x \in [a, b]$ and any open neighborhood $(x - \epsilon, x + \epsilon)$ around x for $\epsilon > 0$. The intersection $(x - \epsilon, x + \epsilon) \cap [a, b]$ always contains infinitely many distinct real numbers. Thus, every point of $[a, b]$ is a limit point of $[a, b]$.

Since $[a, b]$ is closed and every point in $[a, b]$ is a limit point, $[a, b]$ is a nonempty perfect set in $\mathbb{R}^1$.

Step 2: Apply the main theorem.

By the above theorem (regarding perfect sets in $\mathbb{R}^k$), every nonempty perfect set in $\mathbb{R}^k$ is uncountable. Therefore, the interval $[a, b]$ is uncountable.

Step 3: Show $\mathbb{R}$ is uncountable.

Since the real line $\mathbb{R}$ contains the interval $[a, b]$ as a subset, and $[a, b]$ is uncountable, the set of all real numbers $\mathbb{R}$ must also be uncountable. $\square$

2.44 The Cantor Set

The set constructed here shows that there exist nonempty, uncountable perfect sets in $\mathbb{R}^1$ that contain no open segments (intervals).

Step 1: Construction of the Cantor Set

1. Let E_0 be the closed unit interval $[0, 1]$.
2. Remove the open middle third interval $(\frac{1}{3}, \frac{2}{3})$, leaving the set E_1 , which is the union of two closed intervals:

$$E_1 = \left[0, \frac{1}{3}\right] \cup \left[\frac{2}{3}, 1\right]$$

3. Remove the open middle third of each of those remaining two intervals. This gives E_2 , which is the union of $2^2 = 4$ closed intervals:

$$E_2 = \left[0, \frac{1}{9}\right] \cup \left[\frac{2}{9}, \frac{1}{3}\right] \cup \left[\frac{2}{3}, \frac{7}{9}\right] \cup \left[\frac{8}{9}, 1\right]$$

4. Continue this process inductively. At the n -th step, we obtain a set E_n with the following properties:
 - (a) $E_0 \supset E_1 \supset E_2 \supset E_3 \supset \dots$
 - (b) E_n is the union of 2^n disjoint closed intervals, each having length 3^{-n} .

The **Cantor set** P is defined as the infinite intersection of all these sets:

$$P = \bigcap_{n=1}^{\infty} E_n$$

Step 2: Non-Emptiness and Compactness

1. **Non-Empty:** Each E_n is a non-empty closed set. Because $E_n \supset E_{n+1}$ for all n , $\{E_n\}$ is a nested sequence of non-empty compact sets. By the above theorem (regarding nested compact sets), their intersection $P = \bigcap_{n=1}^{\infty} E_n$ is non-empty.
2. **Compact:** Each set E_n is closed and bounded in $\mathbb{R}^1$. Since P is the intersection of closed sets, P is closed. Furthermore, $P \subset E_0 = [0, 1]$, so P is bounded. Because P is a closed and bounded subset of $\mathbb{R}^1$, P is compact.

Step 3: The Cantor Set Contains No Open Interval

By construction, no segment (open interval) of the form

$$\left(\frac{3k+1}{3^m}, \frac{3k+2}{3^m}\right)$$

(where k and m are positive integers) can contain any point of P , because all such middle-third segments were removed during step m .

Now, take any arbitrary open segment (α, β) with $\beta > \alpha$. Since the lengths 3^{-m} shrink to zero as $m \rightarrow \infty$, we can choose an integer m large enough so that:

$$3^{-m} < \frac{\beta - \alpha}{6}$$

Because the removed middle-third segments of length 3^{-m} are spaced at distances no larger than $6 \cdot 3^{-m}$ apart, the interval (α, β) is long enough that it must contain at least one complete removed segment of the form $(\frac{3k+1}{3^m}, \frac{3k+2}{3^m})$.

Since that removed segment contains no points of P , the larger segment (α, β) cannot be entirely contained in P . Thus, P **contains no open interval**.

Step 4: Every Point of P is a Limit Point

To show that P is perfect, it remains to prove that P contains no isolated points.

1. Take any arbitrary point $x \in P$, and let $S = (x - \epsilon, x + \epsilon)$ be any open segment containing x for $\epsilon > 0$.
2. For each stage n , $x \in P \subset E_n$. Let I_n be the specific closed interval of E_n that contains x .
3. The length of I_n is 3^{-n} . Choose n large enough so that $3^{-n} < \epsilon$. This guarantees that $I_n \subset S$.
4. The interval I_n has two endpoints. Let x_n be an endpoint of I_n such that $x_n \neq x$.
5. By the construction of the Cantor set, the endpoints of all sub-intervals at every stage n are never removed, so $x_n \in P$.
6. Thus, in any neighborhood S of x , we have found a point $x_n \in P$ with $x_n \neq x$.

Therefore, **every point of P is a limit point of P** .

Step 5: Conclusion - P is Perfect and Uncountable

Since P is closed and every point of P is a limit point of P , P is a **perfect set**. By Theorem 2.43 (which states that every non-empty perfect set in $\mathbb{R}^k$ is uncountable), it immediately follows that **the Cantor set P is uncountable**.

Remark: Total Length and Lebesgue Outer Measure

Although P is uncountable, its total "length" is zero:

- At stage 1, we remove 1 interval of length $\frac{1}{3}$.
- At stage 2, we remove 2 intervals of length $\frac{1}{9}$.
- At stage n , we remove 2^{n-1} intervals of length $\frac{1}{3^n}$.

Summing the lengths of all removed intervals yields a geometric series:

$$\sum_{n=1}^{\infty} 2^{n-1} \cdot \frac{1}{3^n} = \frac{1}{3} \sum_{n=0}^{\infty} \left(\frac{2}{3}\right)^n = \frac{1}{3} \left(\frac{1}{1 - \frac{2}{3}}\right) = \frac{1}{3} \cdot 3 = 1$$

Since the original interval $[0, 1]$ has a total length of 1, and we removed open intervals whose total lengths sum to 1, the remaining set P has total length zero. In Measure Theory, this is expressed rigorously by stating that **the Cantor set P has Lebesgue outer measure zero** ($\mu(P) = 0$).

Connected Sets

2.45 Definition

Two subsets A and B of a metric space X are said to be **separated** if both $A \cap \overline{B}$ and $\overline{A} \cap B$ are empty, i.e., if no point of A lies in the closure of B and no point of B lies in the closure of A .

A set $E \subset X$ is said to be **connected** if E is not the union of two nonempty separated sets.

2.46 Remark

Separated sets are naturally disjoint, but disjoint sets are not necessarily separated:

- **Disjoint but NOT separated:** Consider the closed interval $[0, 1]$ and the open segment $(1, 2)$. Although they do not share any points, they are not separated because $1 \in [0, 1]$ and 1 is a limit point of $(1, 2)$ (meaning $1 \in \overline{(1, 2)}$). Thus, $[0, 1] \cap \overline{(1, 2)} \neq \emptyset$.

- **Separated sets:** Consider the open segments $(0, 1)$ and $(1, 2)$. The closure of $(0, 1)$ is $[0, 1]$ and the closure of $(1, 2)$ is $[1, 2]$. Neither set contains any point from the closure of the other:

$$(0, 1) \cap [1, 2] = \emptyset \quad \text{and} \quad [0, 1] \cap (1, 2) = \emptyset$$

The connected subsets of the real line $\mathbb{R}^1$ have a particularly simple characterization, which we will establish in the following theorem.

Theorem 1.75. *A subset E of the real line $\mathbb{R}^1$ is connected if and only if it has the following property: If $x \in E$, $y \in E$, and $x < z < y$, then $z \in E$.*

Proof. We prove both directions of the logical equivalence step-by-step.

Step 1: Forward Direction ($\implies$)

We assume E is connected. We must show that for any $x, y \in E$ with $x < z < y$, the intermediate point z must also belong to E .

Suppose, for the sake of contradiction, that there exist points $x \in E$ and $y \in E$ with $x < y$, and some point $z \in (x, y)$ such that $z \notin E$.

We use this point z to decompose E into two sets:

$$A_z = E \cap (-\infty, z) \quad \text{and} \quad B_z = E \cap (z, \infty)$$

Let us examine the properties of A_z and B_z :

- (i) **Non-empty:** Since $x \in E$ and $x < z$, we have $x \in A_z$, so $A_z \neq \emptyset$. Similarly, since $y \in E$ and $y > z$, we have $y \in B_z$, so $B_z \neq \emptyset$.
- (ii) **Union is E :** Since $z \notin E$, every point in E is either strictly less than z or strictly greater than z . Thus, $E = A_z \cup B_z$.
- (iii) **Separated:** Since $A_z \subset (-\infty, z)$, its closure satisfies $\overline{A_z} \subset (-\infty, z]$. Therefore, $\overline{A_z} \cap B_z \subset (-\infty, z] \cap (z, \infty) = \emptyset$. Similarly, since $B_z \subset (z, \infty)$, its closure satisfies $\overline{B_z} \subset [z, \infty)$, which implies $A_z \cap \overline{B_z} \subset (-\infty, z) \cap [z, \infty) = \emptyset$.

Thus, E is the union of two non-empty separated sets A_z and B_z , which means E is not connected. This directly contradicts our initial assumption that E is connected.

Therefore, if E is connected, then for all $x, y \in E$ and $x < z < y$, we must have $z \in E$.

Step 2: Backward Direction ($\impliedby$)

Now we assume E possesses the property that whenever $x, y \in E$ and $x < z < y$, then $z \in E$. We must prove that E is connected.

Suppose, for the sake of contradiction, that E is not connected. Then E can be written as the union of two non-empty separated sets:

$$E = A \cup B, \quad \text{where } A \neq \emptyset, B \neq \emptyset, A \cap \overline{B} = \emptyset, \text{ and } \overline{A} \cap B = \emptyset$$

Pick points $x \in A$ and $y \in B$. Without loss of generality, assume $x < y$. Define the set:

$$A \cap [x, y]$$

This set is non-empty (since $x \in A \cap [x, y]$) and bounded above by y . Therefore, by the Least Upper Bound Property of $\mathbb{R}$, its supremum exists. Let:

$$z = \sup(A \cap [x, y])$$

Since $x \leq z \leq y$, z is a limit point or an element of $A \cap [x, y]$. By the above theorem (regarding properties of closures), $z \in \overline{A}$.

Because A and B are separated, $\overline{A} \cap B = \emptyset$. Since $z \in \overline{A}$, it follows that $z \notin B$. In particular, since $y \in B$, we know $z \neq y$, so $x \leq z < y$.

Now we analyze two possible cases for z :

- **Case 1: $z \notin A$.**

Since $z \notin A$ and $z \notin B$, z does not belong to $A \cup B = E$. Thus, $x \in E$, $y \in E$, and $x < z < y$, but $z \notin E$. This contradicts our hypothesis.

• **Case 2:** $z \in A$.

Since $A \cap \overline{B} = \emptyset$ and $z \in A$, we must have $z \notin \overline{B}$. Because $z \notin \overline{B}$, z is not in B and is not a limit point of B . Therefore, there exists a neighborhood around z that contains no points of B . Specifically, we can find a point z_1 such that $z < z_1 < y$ and the interval $(z, z_1]$ contains no points of B :

$$(z, z_1] \cap B = \emptyset$$

Furthermore, since z is the supremum of $A \cap [x, y]$ and $z_1 > z$, there can be no points of A in $(z, z_1]$ either, so $(z, z_1] \cap A = \emptyset$.

Combining these two facts, $z_1 \notin A$ and $z_1 \notin B$, which implies $z_1 \notin A \cup B = E$. Thus, $x \in E$, $y \in E$, and $x < z_1 < y$, but $z_1 \notin E$. This again contradicts our hypothesis.

Since both cases yield a contradiction, our assumption that E is not connected must be false. Hence, E is connected. $\square$

Remark 1.76. 1. **Connected Subsets of the Real Line are Intervals:**

The property " $x \in E, y \in E, x < z < y \implies z \in E$ " is precisely the defining characteristic of an interval in $\mathbb{R}^1$. Therefore, the connected subsets of $\mathbb{R}^1$ are exactly intervals of every kind: open intervals (a, b) , closed intervals $[a, b]$, half-open intervals $[a, b)$ or $(a, b]$, unbounded rays $(-\infty, a)$, $[a, \infty)$, single points, and the entire line $\mathbb{R}^1$.

2. **Connectedness vs. Path-Connectedness (Topological Intuition):**

In general metric and topological spaces, connectedness mathematically means that a set cannot be split into separated pieces.

In higher topology, you will encounter a related concept called **path-connectedness**, where a space is path-connected if any two points can be joined by a continuous path/curve. While every path-connected set is connected, the converse is not true in general topological spaces—there exist sets that are connected but not path-connected (such as the Topologist's Sine Curve).

This distinction highlights that topological connectedness is fundamentally about the absence of separation or gaps, rather than the existence of geometric paths.